

RESISTOR SORTER AND RETRIEVER

Mini Project Report

Submitted to the APJ Abdul Kalam Technological University

in partial fulfillment of requirements for the award of degree

Bachelor of Technology

in

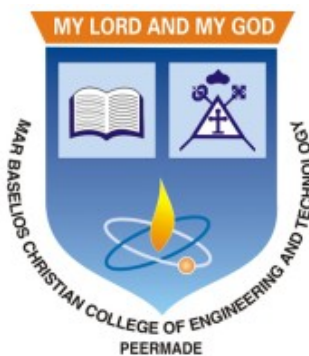
Electronics and Communication Engineering

by

JOSEPH JOHN(MBC23EC025)

JUBIT A JACOB(MBC23EC026)

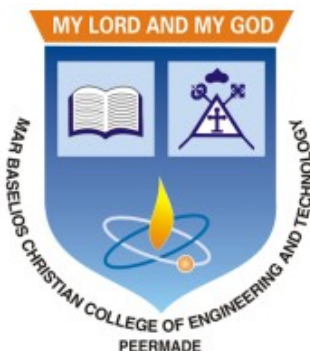
NOUFAL N(MBC23EC030)



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND
TECHNOLOGY, PEERMADE
MARCH 2026

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND
TECHNOLOGY

2025-2026



CERTIFICATE

This is to certify that this report entitled “**RESISTOR SORTER AND RETRIEVER**” submitted by **JOSEPH JOHN (MBC23EC025)**, **JUBIT A JACOB (MBC23EC026)**, **NOUFAL N (MBC23EC030)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of B.Tech. in Electronics and Communication Engineering, is a bonafide report of the Mini Project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Prof. Chinsu K Vijayan
Project Guide
Assistant Professor
Department of ECE

Prof. Tom J Kuriakose
Project Coordinator
Associate Professor
Department of ECE

Dr. Oommen Tharakan K.T
Head of the Department
Professor
Department of ECE

ABSTRACT

The Resistor Sorter and Retriever is an automated system developed to simplify the identification, classification, and storage of resistors in electronics laboratories. The system employs Wheatstone bridge measurement technology to accurately determine the resistance value of an unknown resistor. Three precision resistors with known values are incorporated into the bridge circuit, and the unknown resistance is calculated by analyzing the voltage imbalance produced in the network. An ESP32 microcontroller processes the measured data and determines the exact resistance range of the component. Based on the identified value, a motorized mechanism automatically sorts the resistor into its designated storage compartment. In addition to sorting, the system provides an intelligent retrieval feature that allows users to request and obtain a specific resistor value when required. This dual functionality reduces the time and effort involved in manual resistor identification and searching. The automation significantly minimizes human error and improves component organization. The system is designed to be cost-effective, compact, and suitable for educational and laboratory environments. Overall, the project demonstrates a reliable and efficient solution for automated resistor management, combining both sorting and retrieval in a single integrated platform..

ACKNOWLEDGEMENT

We take this opportunity to express our deepest sense of gratitude and sincere thanks to everyone who helped us to complete this work successfully. We express our sincere thanks to **Dr.V I George**, Principal, Mar Baselios Christian College of Engineering and Technology, Peermade, for his valuable support. We would also like to thank **Dr. Oommen Tharakan K T**, Professor and Head, Department of Electronics and Communication Engineering, for providing all the necessary facilities and support. We would like to express our sincere gratitude to our project coordinator **Prof. Tom J Kuriakose**, Associate Professor Department of Electronics and Communication Engineering for his support and coordination. We would like to express our sincere gratitude to our project guide **Prof. Chinsu K Vijayan**, Assistant Professor, for the guidance throughout the project. We thank our family and friends who contributed to the successful fulfillment of this project work. Finally, we thank the Almighty for the blessings and guidance that enable us to carry out complete this Mini project work successfully

JOSEPH JOHN

JUBIT A JACOB

NOUFAL N

CONTENTS

<i>Abstract</i>	i
<i>Acknowledgement</i>	ii
<i>Contents</i>	iii
<i>List of Figures</i>	iv
<i>Abbreviations</i>	v
1 INTRODUCTION	
1.1 General Background	1
1.2 Limitations of Existing Solutions	1
1.3 Proposed System	2
1.4 Objective	3
2 LITERATURE SURVEY	5
3 BLOCK DIAGRAM	
3.1 Block Diagram	7
3.2 Functional Block explanation	8
3.2.1 The Sensing and Acquisition block	8
3.2.2 The Processing and logic Blocks.....	8
3.2.3 The Actuation and Drive Blocks	9
3.3 System Overview	9
4 HARDWARE REQUIREMENTS	
4.1 Circuit Diagram	10
4.2 Circuit Diagram Explanation and Pin Configurations	11
4.2.1 The ESP32 Dev Module(Central Processing Unit)	11
4.2.2 High-Precision Measurement Block(Wheatstone Bridge and ADC)	11
4.2.3 Motion Control Block(Stepper and Drivers)	12
4.3 Circuit Methodology	12

5	SOFTWARE REQUIREMENTS		
5.1	Arduino IDE		14
5.2	Embedded-C		14
5.3	EasyEDA		15
5.4	ESP-IDF		16
5.5	Flowgraph		17
6	WORKING OF THE SYSTEM		21
7	RESULTS		23
8	PROJECT FEASIBILITY		24
8.1	Sustainable Development Goal(SDG) Alignment		24
9	DISCUSSIONS		
9.1	Advantages		27
9.1.1	Improves Efficiency in Sorting Electronic Components		27
9.1.2	Provides High Accuracy in Resistor Value Detection		27
9.1.3	Enables Automated Sorting and Retrieval		27
9.1.4	Ensures Precise Positioning using Stepper Motor Control		27
9.1.5	Reduces Human Error and Improves Reliability		28
9.1.6	Compact and Cost effective Solution		28
9.2	Disadvantages		28
9.2.1	Initial Setup and Calibration Complexity		28
9.2.2	Mechanical Wear and Maintenance		28
9.3	Applications		28
9.3.1	Electronic Laboratories and Educational Institutions		28
9.3.2	Electronic Manufacturing and Assembly Units		28
9.3.3	Repair and Maintenance Workshop		29
9.3.4	Inventory Management System		29
9.3.5	hobbyist and DIY Electronic Projects		29
9.4	Future Scope		29

10 CONCLUSIONS	31
REFERENCES	33
APPENDICES	34
APPENDIX 1:BASE PAPER	35
APPENDIX 2:CODE	40
APPENDIX 3:DATA SHEETS	48

LIST OF FIGURES

FIG 3.1	Block Diagram of the system	7
FIG 4.1	Circuit Diagram of the system	10
FIG 5.5	Flowgraph of the System	17
FIG 7.1	Figure of Resistor Sorter and Retriever System	23
FIG 7.2	Figure of Resistor Sorter and Retriever System	23
TABLE 8.1	SDG Mapping of the System	24

ABBREVIATIONS

ADC - Analog to Digital Converter
ADS1115 - 16-bit Analog to Digital Converter Module
DC - Direct Current
DMM - Digital Multimeter
ESP32 - Espressif 32-bit Microcontroller
GUI - Graphical User Interface
I2C - Inter-Integrated Circuit Communication Protocol
IDE - Integrated Development Environment
IoT - Internet of Things
LM2596 - Step-Down Buck Converter Module
PWM - Pulse Width Modulation
RSR - Resistor Sorter and Retriever
SDG - Sustainable Development Goals
SMPS - Switched Mode Power Supply
TB6600 -Stepper Motor Driver Module
VCC - Voltage at the Common Collector
GND - Ground
Wi-Fi - Wireless Fidelity
EasyEDA - Electronic Design Automation
ESP-IDF - Espressif IoT Development Framework

CHAPTER 1

INTRODUCTION

1.1 General Background

In the field of electronic engineering, the resistor is one of the most essential and universally used passive components. From simple educational experiments to advanced industrial electronics, resistors play a vital role in controlling current flow, dividing voltages, biasing transistors, and protecting sensitive circuit elements. As modern electronic systems continue to grow in complexity and scale, laboratories, educational institutions, and manufacturing facilities are required to maintain extensive inventories of resistors with different resistance values, tolerances, wattage ratings, and packaging types.

Managing such a large collection of components efficiently has become a significant challenge. Traditionally, resistors are identified using the standardized color code system, which requires visual interpretation of colored bands printed on the resistor body. While this method is effective for small-scale work, it becomes increasingly inefficient and error-prone when dealing with large quantities of resistors. Another common method involves manually measuring each resistor using a digital multimeter (DMM), which, although accurate, is highly time-consuming.

As the demand for rapid prototyping and fast-paced product development increases, efficient component management has become crucial. Poor organization of resistors can lead to workflow delays, increased project time, and potential circuit failures due to incorrect component selection. Therefore, there is a growing need for automated solutions that can accurately identify, sort, store, and retrieve resistors in a systematic and reliable manner.

1.2 Limitations of Existing Solutions

Conventional resistor sorting and retrieval methods rely heavily on manual human intervention. Although these methods are widely practiced, they present several critical limitations that negatively affect efficiency and reliability.

- **Human Error:** Manual interpretation of resistor color bands is susceptible to mistakes. Factors such as poor lighting conditions, faded color bands, color blindness, and the presence of resistors with similar color shades can lead to incorrect value identification. Installing the wrong resistor in a circuit may result in malfunction, component damage, or inaccurate experimental results.

- **Time Inefficiency:** Searching through mixed or poorly labeled resistor storage boxes is a slow and tedious process. In many laboratories, engineers and students spend a significant amount of time measuring resistors individually before use. This repetitive task reduces the time available for actual design, analysis, and testing activities.
- **Physical Strain and Fatigue:** Manual sorting of tiny electronic components requires continuous visual focus and repetitive hand movements. Over time, this leads to eye strain, reduced concentration, and physical fatigue, which further increases the likelihood of mistakes.
- **Scalability and Inventory Challenges:** Traditional storage systems lack the ability to provide real-time tracking of available components. As the number of stored resistors increases, maintaining an organized inventory becomes difficult. Manual systems are not suitable for large-scale laboratories or production environments where quick access to specific resistor values is essential.

These limitations highlight the necessity of developing an automated system capable of improving efficiency, accuracy, and scalability.

1.3 Proposed System

To address the shortcomings of conventional methods, this project proposes the design and development of Resistor Sorter and Retriever, an electromechanical system that integrates electronics, mechanical design, and embedded programming.

The system employs the microcontroller platform; ESP32 as the central control unit. A precision resistance measurement circuit—implemented using a Wheatstone bridge configuration—is used to accurately determine the resistance value of each inserted resistor. The measured value is processed by the microcontroller and classified into predefined resistance ranges.

A motorized mechanical assembly driven by servo motors performs the physical sorting operation. Based on the measured value, the system automatically directs the resistor into the appropriate storage compartment. This eliminates the need for manual classification and significantly reduces sorting time.

In addition to sorting, the system incorporates an intelligent retrieval mechanism. Users can request a specific resistor value through a user interface such as push buttons and a display module. Upon receiving the request, the system locates the corresponding storage bin and the user can take the required resistor.

By combining electronic sensing, embedded control, and mechanical automation, the proposed system provides a reliable, efficient, and user-friendly solution for resistor inventory management.

1.4 Objective

The main objective of the Resistor Sorter and Retriever is to design and develop a smart system capable of automatically identifying, sorting, and retrieving resistors in an efficient and reliable manner. The specific objectives of the project are listed below:

- To design an automated system capable of accurately measuring the resistance value of resistors using an electronic measurement circuit.
- To develop a microcontroller-based control system using ESP32 for processing measurement data and controlling the sorting mechanism.
- To implement an automated sorting mechanism using servo motors and a stepper motor for precise positioning and movement.
- To reduce manual effort and minimize human errors in resistor identification and storage.
- To design an organized storage and retrieval mechanism for efficient management of electronic components in laboratories.
- To provide a user-friendly interface using an OLED display and push buttons for system monitoring and control.
- To improve laboratory productivity by reducing the time required for resistor identification and searching.
- To develop a cost-effective and scalable prototype suitable for educational institutions and small electronics workshops.
- To demonstrate the practical application of embedded systems, automation, and mechatronics in solving real-world engineering problems.

The remainder of this report is organized as follows. Chapter 2 presents the literature survey of existing automated component identification and sorting systems. Chapter 3 describes the block diagram of the proposed Resistor Sorter and Retriever along with the functional explanation of sensing, processing, display, and actuation modules. Chapter 4 explains the hardware requirements and circuit connections of the ESP32 microcontroller, Wheatstone bridge measurement circuit, servo motors, stepper motor driver, and OLED display. Chapter 5 discusses the software requirements used for implementation, including the Arduino IDE and embedded programming techniques. Chapter 6 explains the working of the proposed system in detail. Chapter 7 presents the results obtained from the implementation and testing of the system. Chapter 8 discusses the project feasibility,

including Sustainable Development Goals (SDG) alignment and cost estimation. Finally, the references used for the development of this project are listed at the end of the report.

CHAPTER 2

LITERATURE SURVEY

This chapter presents a detailed review of existing research works related to automated sorting systems, embedded control mechanisms, and intelligent classification techniques. These studies provide essential insights into the design and development of the proposed **Resistor Sorter and Retrivaler**, particularly in areas such as sensing, decision-making, and actuation. The strengths and limitations of each approach are analyzed to justify the design choices of the proposed system.

Automated sorting using microcontroller-based systems has been widely explored due to its simplicity and cost-effectiveness. A study on an Arduino-based automated sorting and shredding system demonstrates the integration of embedded control with mechanical actuation to perform sorting tasks efficiently [3]. The system utilizes Arduino for real-time control and decision-making, making it suitable for low-cost automation applications. One of the major advantages of this approach is its simplicity, ease of implementation, and adaptability to various sorting scenarios. However, the system is designed for general object sorting and does not provide mechanisms for identifying specific properties such as resistor values or color bands. Additionally, its reliance on basic sensing techniques limits accuracy when dealing with small and visually similar components like resistors.

Machine vision-based sorting systems represent a more advanced approach by incorporating image processing and pattern recognition techniques. The study on automatic sorting systems based on machine vision highlights the use of cameras and image analysis algorithms to classify objects based on visual features [2]. This method is highly advantageous for applications requiring precise identification, such as recognizing resistor color codes. It enables high accuracy and flexibility in classification tasks. However, the major drawback of this approach lies in its computational complexity and high processing requirements. The need for powerful processors, controlled lighting conditions, and sophisticated algorithms increases system cost and reduces feasibility for compact embedded implementations. Furthermore, real-time performance may be affected in resource-constrained environments.

Infrared (IR) sensor-based sorting systems provide a simpler and more practical alternative for object detection. An experimental study on an automatic sorting machine using IR proximity sensors and microcontrollers demonstrates effective detection and actuation mechanisms [4]. The system detects the presence of objects and triggers sorting actions accordingly, making it suitable for basic automation tasks. Its advantages include low

cost, ease of integration, and reliable performance for object detection. However, IR sensors are limited to detecting object presence and cannot identify detailed characteristics such as resistance value or color coding. This limitation makes the approach unsuitable for applications like resistor sorting, where precise classification is essential.

Recent developments in intelligent sorting systems have introduced the use of deep learning and advanced imaging techniques. A study on an intelligent robotic sorting system using RGB-D imaging and deep learning algorithms demonstrates highly accurate object classification in real time [1]. The system leverages depth information and neural networks to improve classification accuracy and adaptability to different objects. While this approach offers superior performance and scalability, it suffers from significant disadvantages, including high computational requirements, increased system complexity, and the need for extensive training datasets. These factors make it impractical for low-cost and portable systems such as the proposed resistor sorter.

A comprehensive survey on hardware-based sorting architectures provides an overview of various sorting techniques and their efficiencies [5]. The study compares traditional sorting mechanisms with modern hardware implementations, offering insights into optimizing sorting performance. The primary advantage of this work lies in its broad analysis of sorting methodologies and design considerations. However, as a generalized survey, it does not address application-specific challenges, particularly those related to sorting small electronic components like resistors. Additionally, it lacks practical implementation details that can be directly applied to embedded systems.

From the above literature review, it is evident that existing systems either focus on general-purpose sorting or rely on computationally intensive techniques. Microcontroller-based systems are simple and cost-effective but lack precision, while machine vision and deep learning approaches provide high accuracy at the expense of complexity and cost. Sensor-based systems offer practical implementation but are limited in classification capability. Therefore, there exists a research gap in developing a system that combines **low cost, simplicity, and accurate resistor identification**.

The proposed **Resistor Sorter and Retriever** aims to bridge this gap by integrating embedded system design with efficient sensing and sorting mechanisms tailored specifically for resistors. By optimizing between performance and cost, the system ensures practical applicability in educational environments, laboratories, and small-scale industries.

CHAPTER 3

BLOCK DIAGRAM

3.1 Block Diagram

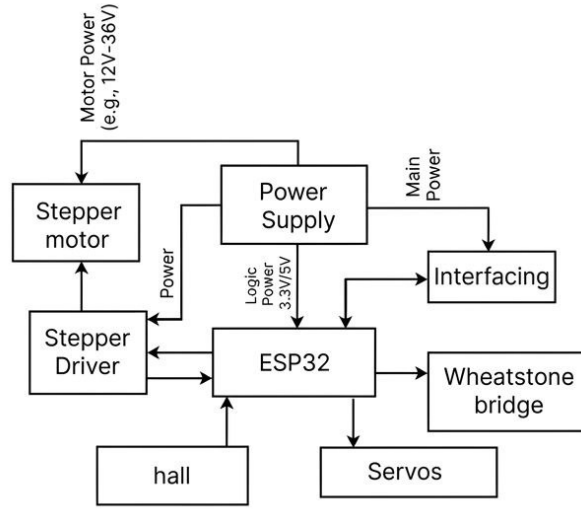


Fig 3.1:Block Diagram of the system

Fig 3.1 illustrates the block diagram of Resistor Sorter and Retriever (RSR) is an automated electromechanical system, designed to bridge the gap between manual component management and industrial-grade precision. In the field of electronics engineering, the organization of passive components like resistors is often a tedious and error-prone task. This project introduces a "Sense-Process-Act" framework to automate the identification and storage of resistors. The system is architected to perform three primary functions: high-resolution resistance measurement, intelligent data categorization, and precision mechanical placement. Unlike traditional sorters that may rely on simple voltage dividers, this system utilizes a Bridge-ADC configuration to ensure laboratory-grade accuracy across a wide range of resistance values. The above diagram illustrates the interconnected nature of the system. The flow starts with the physical interface (the

measurement bridge), moves through the digital processing unit (ESP32), and concludes with the mechanical drive system (Stepper and Servo motors).

3.2 Functional Block Explanation

- **3.2.1 The Sensing and Acquisition Block:**

This block serves as the sensory organ of the project, converting the physical properties of a resistor into a digital format.

- **Wheatstone Bridge Circuit:** The core of the measurement system is a classic Wheatstone bridge. By placing the unknown resistor in a balanced bridge configuration, the system produces a differential voltage proportional to the resistance value. This method is preferred over a standard voltage divider because it is less susceptible to power supply fluctuations and offers much higher sensitivity for low-tolerance resistors.

- **ADS1115 (16-bit High-Resolution ADC):** The analog signal from the bridge is too minute for the standard pins of a microcontroller. The ADS1115 acts as an external Analog-to-Digital Converter. With its 16-bit resolution, it can detect voltage changes as small as 188 microvolts. It communicates with the ESP32 via the I2C protocol, providing "clean," high-fidelity data that is essential for distinguishing between resistors of similar color codes (e.g., 220Ω vs 330Ω).

- **Hall Effect Sensor:** This is a magnetic sensor used for mechanical calibration. It provides a "Home" reference signal. When the mechanical arm or tray reaches its starting position, the Hall sensor triggers a signal to the ESP32, ensuring that the system's coordinate system is reset to zero before every sorting cycle.

- **3.2.2 The Processing and Logic Block:**

This block acts as the "Brain," coordinating all subsystems and performing the mathematical heavy lifting.

- **ESP32 Dev Module:** The ESP32 is a powerful dual-core microcontroller chosen for its high clock speed and multitasking capabilities.

- **Core 0:** Dedicated to high-speed data acquisition from the ADS1115 and calculating the resistance using the bridge equations.

- **Core 1:** Manages the Pulse Width Modulation (PWM) signals for the servo motors and the step-timing for the stepper motor.

- **System Firmware and Logic:** The software implements a sorting algorithm. Once the resistance is calculated, the system classifies it into pre-defined "Bins" (e.g., $3.3k\Omega$, $10k\Omega$, $12k\Omega$, $100k\Omega$, etc.). The logic then determines the exact rotational angle or linear distance the mechanical system must travel to reach that bin.

- **3.2.3 The Actuation and Drive Block:**

This block converts electrical commands into physical force and movement.

- **TB6600 Stepper Driver:** Stepper motors require high current and precise timing. The TB6600 is an industrial-grade driver that handles the power delivery to the motor. It protects the ESP32 from high voltage and allows for "micro-stepping," which provides the system with sub-millimeter positioning accuracy.

- **Nema 17 Stepper Motor:** This is the primary actuator for the "Retrieving" mechanism. Known for its high holding torque and precise rotational control, the Nema 17 moves the mechanical assembly to the exact storage compartment corresponding to the measured resistor.

- **Servo Motors (3 Units):** While the stepper motor handles the "large" movements, the servos handle the "fine" movements.

3.3 System Overview

The Resistor Sorter and Retriever operates in a closed-loop sequence to ensure maximum reliability. The overall system methodology can be summarized in the following operational phases:

- **Phase 1: Initialization and Homing** Upon power-up, the system enters a self-calibration mode. The Nema 17 motor rotates until the Hall Effect sensor is triggered. This establishes the "Global Zero" of the system, ensuring that any previous mechanical shifts do not affect the sorting accuracy.
- **Phase 2: Measurement and Classification** When a resistor is introduced, the Wheatstone bridge generates an analog voltage. The ADS1115 samples this voltage at high speed, and the ESP32 applies a digital filter to remove ambient electrical noise. The calculated resistance is then categorized into a specific data class.
- **Phase 3: Mechanical Retrieval and Sorting** The ESP32 sends a sequence of pulses to the TB6600 driver, moving the retrieval arm to the bin location. Simultaneously, the servo motors actuate the funnel and mechanical arms to guide the resistor from the measurement base into its designated compartment.
- **Phase 4: Power Management and Safety** The system utilizes a split power architecture. A high-voltage DC supply (12V-24V) powers the mechanical actuators, while a regulated 3.3V/5V rail provides stable power to the sensitive logic and sensing blocks. This isolation is critical to prevent the "back-EMF" from the motors from interfering with the delicate resistance measurements.

CHAPTER 4

HARDWARE REQUIREMENTS

4.1 Circuit Diagram

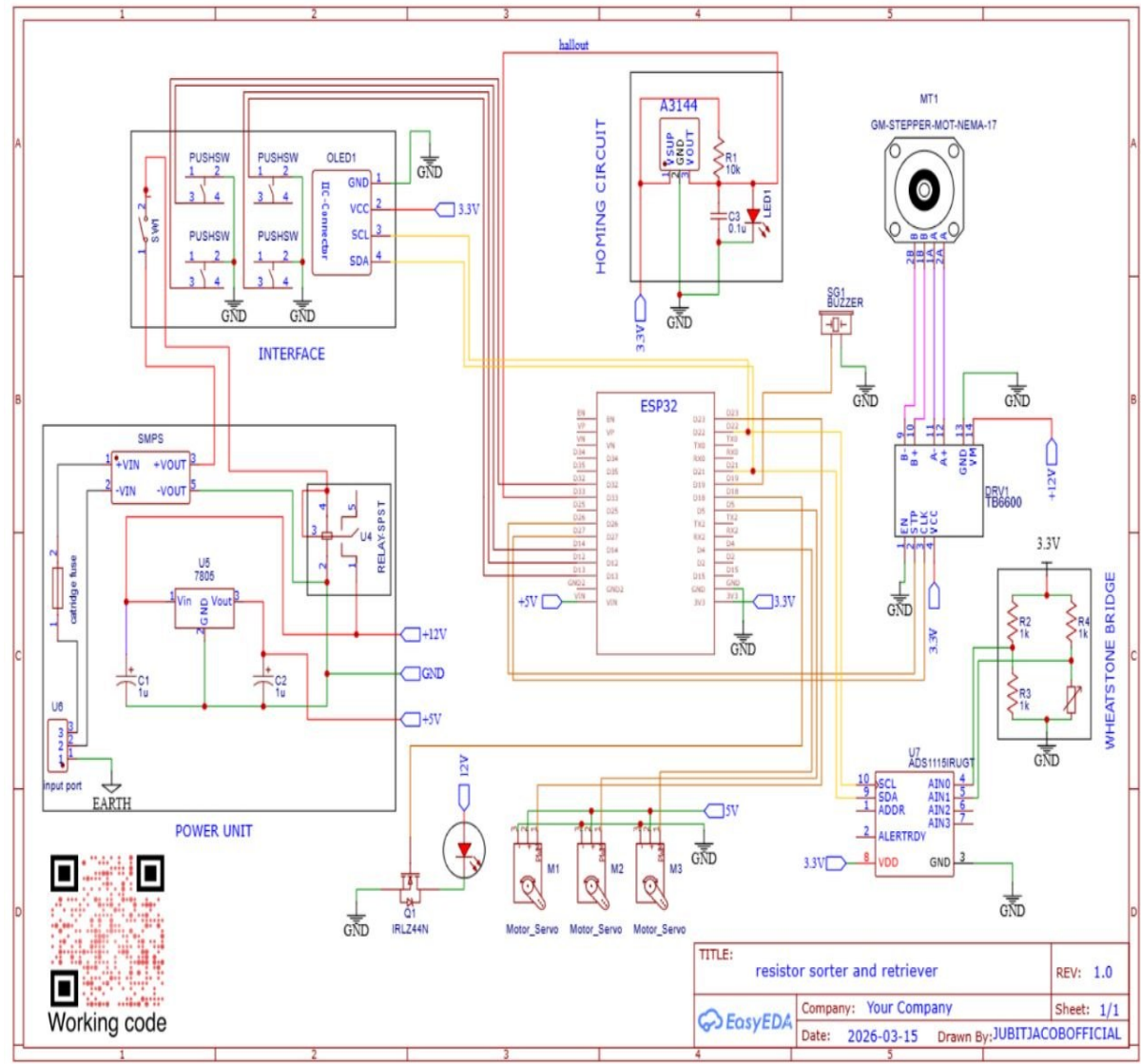


Fig 4.1: Circuit Diagram of the system

Fig 4.1 illustrates circuit design of the Resistor Sorter and Retriever (RSR), which is engineered to provide a stable electrical environment for high-precision analog measurements while managing high-current inductive loads. The circuit acts as the bridge

between the digital logic of the ESP32 and the physical world. The system is powered by a central DC source, which is then distributed through various voltage regulation stages to ensure that the "noise" generated by the motors does not interfere with the sensitive Wheatstone bridge measurements. The schematic ensures signal integrity by utilizing the I2C communication protocol for data-heavy components and dedicated PWM channels for mechanical actuation.

- **4.2 Circuit diagram Explanation and Pin Configurations**

- **4.2.1 The ESP32 Dev Module (Central Processing Unit):** The ESP32 is the core of the circuit. It manages the timing for motor drivers, processes I2C data from the ADC, and drives the user interface.

- **Pin Configuration:**

- VIN / GND:** Connected to the 5V regulated power rail and common ground.

- GPIO 21 (SDA) / GPIO 22 (SCL):** Main I2C bus connected to the ADS1115 and OLED Display.

- GPIO 14 (STEP):** Outputs high-frequency pulses to the TB6600 to control motor speed.

- GPIO 27 (DIR):** Outputs a High/Low signal to the TB6600 to control motor direction.

- GPIO 13:** Input pin for the Hall Effect sensor (Homing signal).

- GPIO 18, 19, 23:** PWM output pins connected to the Signal pins of the three Servo Motors.

- **4.2.2 High-Precision Measurement Block (Wheatstone Bridge and ADC):**

This sub-circuit is responsible for the "Sorting" logic by identifying the resistor value.

- **A. Wheatstone Bridge:**

- Excitation Pins:** Connected to a filtered 3.3V supply to maintain a constant reference.

- Output Nodes:** The differential voltage points are connected to the A0 and A1 pins of the ADS1115.

- **B. ADS1115 (16-bit ADC):**

- VDD / GND:** Connected to 3.3V and Common Ground.

- SDA / SCL:** Connected to the ESP32 I2C bus.

- ADDR:** Tied to GND to set the I2C address to 0x48.

- A0 / A1:** Differential analog inputs from the Wheatstone bridge.

$$V_{out} = V_s \times \left(\frac{R_x}{R_3 + R_x} - \frac{R_2}{R_1 + R_2} \right) \quad (1)$$

- **4.2.3 Motion Control Block (Stepper and Drivers):** This section manages the "Retrieving" mechanism of the project.
 - **A. TB6600 Stepper Driver:**
 - PUL+ / DIR+:** Signal inputs connected to ESP32 GPIO 14 and 27.
 - PUL- / DIR-:** Tied to common Signal Ground.
 - VCC / GND:** Connected to the primary 12V-24V DC power rail.
 - A+/A- and B+/B-:** Output terminals connected to the four wires of the Nema 17 Stepper Motor.
 - **B. Servo Motors (SG90):**
 - Red Wire (VCC):** Connected to the 5V high-current rail.
 - Brown Wire (GND):** Connected to common ground.
 - Orange Wire (PWM):** Connected to the designated ESP32 PWM GPIOs.
- **4.3 Circuit Methodology**

The circuit methodology for the RSR project focuses on Power Decoupling and Signal Precision.

 - **Voltage Regulation:** The system uses a multi-stage approach. An SMPS provides 12V for the Nema 17 motor. An LM2596 Buck Converter steps this down to 5V for the ESP32 and Servo motors. Finally, the ESP32's on-board regulator provides a clean 3.3V for the sensitive ADC and Hall sensor.
 - **Noise Mitigation:** Electrolytic capacitors (1000uF) are placed across the motor power rails to absorb back-EMF spikes, while ceramic capacitors (0.1uF) are placed near the ADS1115 to filter high-frequency noise.
 - **Communication Protocol:** By utilizing I2C (Inter-Integrated Circuit), the system maintains a digital data flow for measurements, which is far more resistant to the electromagnetic interference generated by the stepper motor compared to traditional analog wiring.
 - **Homing Logic:** The Hall Effect sensor circuit provides a fail-safe starting point. When the magnet on the mechanical arm passes the sensor, the circuit pulls the ESP32 pin LOW, triggering an interrupt that resets the internal step-counter to zero.

This circuit architecture ensures that the Resistor Sorter and Retriever can operate continuously with a high degree of measurement accuracy and mechanical repeatability, making it suitable for a professional laboratory environment.

CHAPTER 5

SOFTWARE REQUIREMENTS

- **5.1 Arduino IDE:**

The Arduino Integrated Development Environment (IDE) is the primary software tool used for programming and controlling the microcontroller in the proposed Resistor Sorter and Retriever System. It provides a simple and user-friendly interface for writing, compiling, and uploading programs written in Embedded C/C++ to the Arduino-based controller. The platform supports a wide range of libraries and simplifies communication between hardware components and the microcontroller.

In this project, the Arduino IDE is used to interface various components such as sensors, display units, motors, and sorting mechanisms. The system continuously monitors input signals from the sensing unit, which identifies the resistor value either through measurement or detection techniques. These signals are processed by the Arduino microcontroller and compared with predefined conditions to determine the appropriate sorting action.

When a resistor is detected and its value is identified, the Arduino executes control logic to activate the corresponding sorting mechanism. This may include triggering servo motors or actuators to place the resistor into the correct category. Additionally, the system may display the detected resistor value and sorting status on a display module, improving user interaction and system transparency.

The Arduino IDE also supports serial monitoring, which is useful for debugging and testing during system development. Developers can observe real-time data such as sensor readings and system responses, allowing for efficient troubleshooting. Its extensive library support and ease of use make it highly suitable for embedded automation projects like the proposed system.

- **5.2 Embedded C:**

Embedded C is the programming language used to develop the control logic of the Resistor Sorter and Retriever System. It is a specialized version of the C programming language designed for programming embedded systems and microcontrollers.

Embedded C enables efficient control of hardware components and real-time system operations.

In this project, Embedded C is used to write programs that handle sensor input processing, decision-making, and actuator control. The program continuously reads input data from the resistor detection unit and processes it to determine the resistor's value. Based on the detected value, conditional statements are used to execute the appropriate sorting operation.

The use of Embedded C allows precise control over timing and hardware interactions, which is essential for accurate sorting and retrieval operations. It also enables efficient memory usage and faster execution, making it ideal for microcontroller-based systems. Furthermore, modular programming techniques are used to organize the code into functions such as sensing, processing, and actuation, improving readability and maintainability.

Error handling and validation routines are also implemented using Embedded C to ensure reliable system performance. These include handling incorrect readings, avoiding repeated detections, and ensuring proper synchronization between sensing and sorting operations.

- **5.3 EasyEDA:**

EasyEDA is an online Electronic Design Automation (EDA) tool used for designing and simulating electronic circuits in the proposed system. It provides an integrated environment for schematic design, circuit simulation, and PCB layout, making it a valuable tool during the hardware development phase.

In this project, EasyEDA is used to design the circuit diagram of the Resistor Sorter and Retriever System, including the microcontroller, sensors, motor drivers, power supply, and other supporting components. The platform allows visualization of circuit connections and helps ensure correct interfacing between different modules.

The simulation feature of EasyEDA enables preliminary testing of circuit behavior before physical implementation. By simulating the circuit, potential errors such as incorrect connections, component mismatches, and power issues can be identified and corrected at an early stage. This reduces development time and minimizes

hardware-related issues.

Additionally, EasyEDA supports PCB design, allowing the transition from schematic to a compact and organized printed circuit board layout. This improves system reliability and makes the final prototype more robust and suitable for real-world applications.

Thus, EasyEDA plays a crucial role in validating circuit design, reducing implementation errors, and ensuring the overall efficiency of the system development process.

- **5.4 ESP-IDF (Espressif IoT Development Framework):**

ESP-IDF (Espressif IoT Development Framework) is the official software development framework provided by Espressif Systems for programming the ESP32 microcontroller. It is a powerful and flexible platform that supports development in C and C++ and provides low-level control over the hardware, making it suitable for real-time embedded applications.

In this project, ESP-IDF is used to implement the complete control logic of the resistor sorter and retriever system. It enables precise handling of peripheral interfaces such as GPIO, I2C, and PWM, which are essential for coordinating the operation of sensors, ADC modules, and actuators. The ADS1115 ADC communicates with the ESP32 through the I2C protocol, and ESP-IDF provides robust driver support for reliable data acquisition. The framework is also used to generate accurate timing signals required for controlling the NEMA 17 stepper motor via the TB6600 driver and for positioning the servo motors.

Additionally, ESP-IDF supports multitasking through the FreeRTOS operating system, allowing simultaneous execution of tasks such as resistance measurement, motor control, and user interface updates. This ensures efficient and real-time system performance. The framework also offers debugging, logging, and hardware abstraction features, which enhance development efficiency and system reliability.

Overall, ESP-IDF provides a stable, scalable, and efficient software environment for implementing the automated resistor sorting and retrieval process.

- 5.5 Flowgraph of the System:

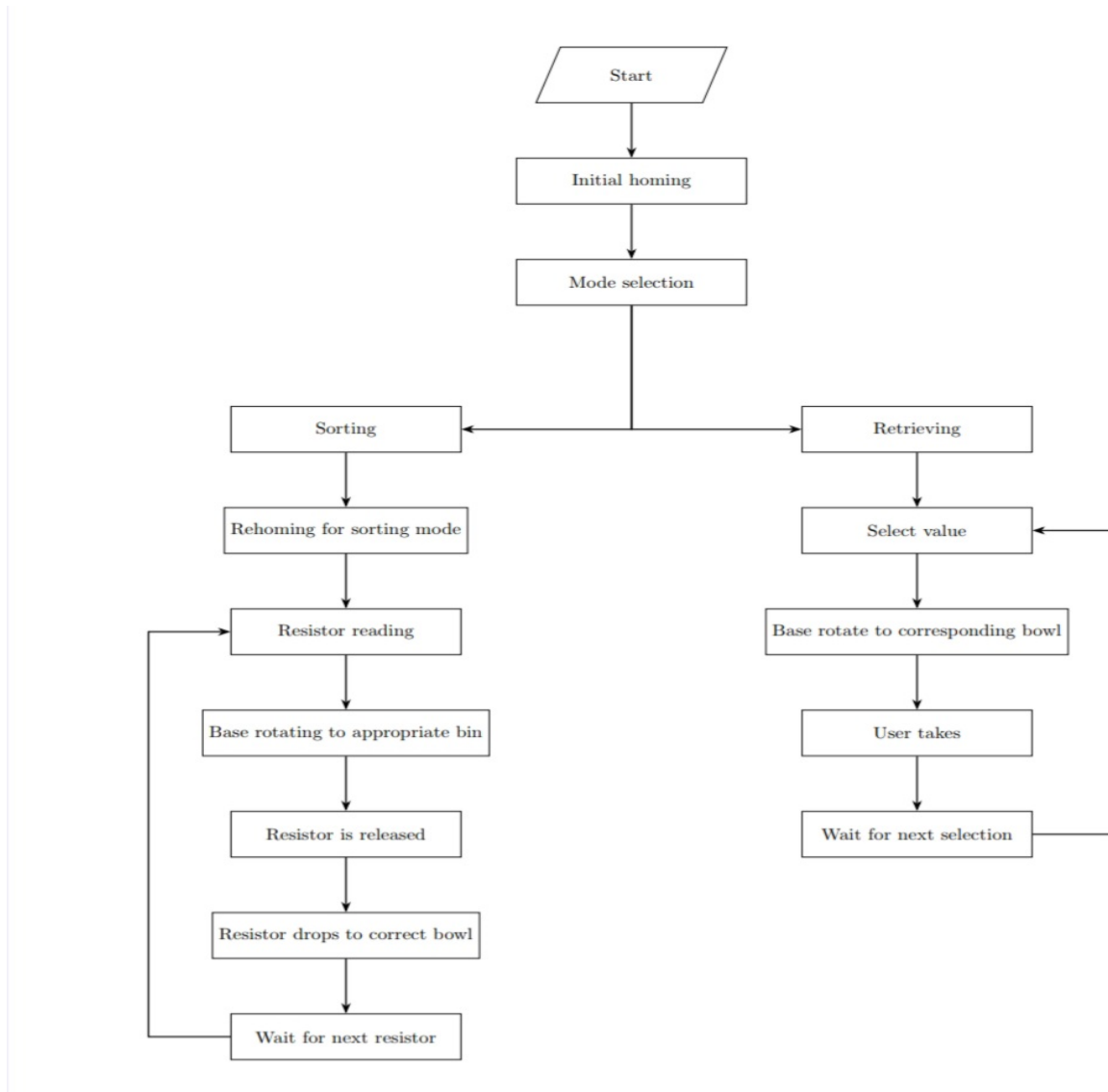


Fig 5.5:Flowgraph of the system

System Operational Logic

The operational logic of the Resistor Sorter and Retriever system follows a dual-mode branching structure initiated after system calibration.

1. System Initialization

The process begins with the **Initial homing** sequence, ensuring all mechanical components are at their reference positions. The user then interacts with the system through the **Mode selection** interface to choose between sorting or retrieving tasks.

2. Sorting Mode

The sorting sequence is designed to identify and organize resistors into specific bins:

- **Rehoming:** The system performs a mode-specific reset to prepare the mechanical arms.
- **Identification:** The system performs a **Resistor reading** to determine the component's value.
- **Mechanical Alignment:** The **Base rotates** to the appropriate bin corresponding to the identified value.
- **Deposition:** The **Resistor is released** and drops into the correct bowl for storage.
- **Automation:** The system enters a **Wait** state before automatically looping back to read the next resistor.

3. Retrieving Mode

The retrieving sequence allows for the automated collection of specific components:

- **Selection:** The user selects the desired resistor value.
- **Retrieval:** The **Base rotates** to the corresponding bowl where that specific value is stored.
- **Collection:** The system provides the component for the user to take.
- **Cycle:** The system waits for the next user selection to repeat the process.
- **Software Description**

Introduction

The software implementation of the Resistor Sorter and Retriever system is designed to ensure accurate, efficient, and automated operation of the entire setup. The system is developed using the ESP-IDF framework for the ESP32 microcontroller, which provides reliable control over hardware peripherals and supports real-time processing.

The software coordinates the interaction between sensing, processing, and actuation components. It reads analog data from the Wheatstone bridge through the ADS1115 ADC, processes the resistance values, and controls motors and actuators accordingly. Additionally, it enables user interaction through push buttons and

provides real-time feedback via the OLED display.

The system operates in two primary modes: Sorting Mode and Retrieving Mode. Based on user input, the software dynamically switches between these modes and executes the corresponding control logic. The inclusion of homing mechanisms and structured workflow ensures precision, repeatability, and efficient operation.

Step-by-Step Working

1. System Initialization

The system is powered ON, and the ESP32 initializes all peripherals such as GPIO pins, I2C communication for ADS1115 and OLED, and PWM signals for motor control.

2. Initial Homing

The stepper motor rotates until the homing sensor (A3144) is triggered. This establishes a reference position for accurate movement of the base.

3. Mode Selection

The user selects the operating mode (Sorting or Retrieving) using push buttons. The system branches into the selected mode.

4. Sorting Mode Operation

In sorting mode, the system first ensures proper alignment through rehomings. A resistor is then placed in the measurement section. The Wheatstone bridge produces a voltage corresponding to the resistance value, which is converted into digital form using the ADS1115 ADC. The ESP32 processes this data and determines the resistance value.

Based on the calculated resistance, the system identifies the appropriate bin. The stepper motor rotates the base to the corresponding position, and the servo motor releases the resistor. The resistor then falls into the correct bin. The system continuously repeats this process for incoming resistors.

5. Retrieving Mode Operation

In retrieving mode, the user selects the required resistance value using input buttons. The ESP32 identifies the corresponding bin and rotates the base to that position using the stepper motor. The user can then manually collect the resistor. The system waits for further user input and continues operation.

6. Continuous Operation

The system operates in a continuous loop, allowing seamless switching between sorting and retrieving modes while maintaining accuracy and efficiency.

CHAPTER 6

WORKING OF THE SYSTEM

The proposed Resistor Sorter and Retriever System operates through the coordinated interaction of sensing, processing, and actuation units controlled by the ESP32 microcontroller. The ESP32 serves as the central control unit, managing data acquisition, decision-making, and control signals for all connected components. It ensures proper synchronization between the Wheatstone bridge sensing circuit, analog-to-digital conversion using the ADS1115 module, and the actuation mechanisms including the stepper motor and servo motors. This centralized control enables accurate and real-time sorting of resistors based on their measured values.

The system begins its operation with the placement of a resistor in the sensing section, where a Wheatstone bridge circuit is used to determine the resistance value. The bridge converts the resistance variation into a small analog voltage signal. This signal is then fed into the ADS1115 analog-to-digital converter, which provides high-resolution digital output to the ESP32. The microcontroller processes this data and calculates the corresponding resistance value using predefined calibration parameters. This method ensures precise measurement even for small resistance variations.

To maintain positional accuracy and repeatability, a homing mechanism is incorporated using the A3144 Hall-effect sensor. At the start of operation, the system performs a homing routine where the stepper motor rotates until the sensor detects a reference position. This establishes a fixed starting point for the sorting mechanism. The NEMA 17 stepper motor, controlled via the TB6600 stepper driver, is then used to rotate the sorting platform or indexing mechanism to specific positions corresponding to different resistor value ranges. The use of a stepper motor ensures precise angular movement and accurate alignment with sorting bins.

Once the resistor value is determined, the ESP32 compares it with predefined categories or ranges. Based on this comparison, the system generates control signals to position the stepper motor at the appropriate sorting location. After positioning, servo motors are activated to perform the sorting action, such as pushing or directing the resistor into the designated compartment. The coordinated movement between the stepper motor and servo motors ensures efficient and accurate sorting with minimal mechanical error.

In addition to sorting, the system also supports retrieval functionality. The stepper motor can rotate the platform to a specific bin location upon user request, allowing easy access to the required resistor value. This enhances the usability of the system, especially in laboratory and educational environments where quick component access is essential.

The integration of high-resolution sensing through the ADS1115, precise motion control using the NEMA 17 stepper motor, and reliable positioning via the homing sensor significantly improves system accuracy and repeatability. The ESP32 continuously monitors all operations and executes control logic in real time, ensuring smooth and efficient system performance.

Overall, the Resistor Sorter and Retriever System provides an effective solution for automated resistor classification and storage. The combination of accurate resistance measurement, precise motor control, and intelligent processing enables the system to operate reliably with minimal human intervention. This integrated approach enhances efficiency, reduces manual effort, and improves organization in electronic component handling.

CHAPTER 7

RESULTS



Fig 7.1:Figure of Resistor Sorter and Retriever system



Fig 7.2:Figure of Resistor Sorter and Retriever system

CHAPTER 8

PROJECT FEASIBILITY

The feasibility of the Automated Resistor Sorter and Retriever is analyzed in terms of Sustainable Development Goals (SDG alignment).

- **8.1 Sustainable Development Goals (SDG) Alignment:**

The development of the Automated Resistor Sorter and Retriever is not only technically significant but also socially and environmentally relevant. The system contributes to several United Nations Sustainable Development Goals (SDGs) by improving productivity, encouraging automation, and promoting responsible use of electronic resources. The project mainly aligns with SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), and SDG 12 (Responsible Consumption and Production).

SDG Number	Definition	Justification
SDG No:- 9	Industry, Innovation and Infrastructure	The resistor sorter and retriever introduces automation into electronics component handling. It enhances accuracy, reduces manual dependency, and supports the development of efficient and innovative laboratory infrastructure.
SDG No:- 12	Responsible Consumption and Production	By enabling correct identification and reuse of resistors, the system minimizes component wastage. It promotes efficient use of electronic resources and supports sustainable production practices.
SDG No:- 8	Decent Work and Economic Growth	The automated system reduces repetitive manual tasks and human errors. It improves productivity in electronics labs and encourages safer and more efficient working environments.

Table 8.1:SDG Mapping of the system

SDG 8: Decent Work and Economic Growth

SDG 8 aims to promote sustained economic growth, productive employment, and safe working environments. In electronics laboratories and small manufacturing environments, resistor identification and sorting are typically performed manually using resistor color codes or handheld multimeters. These traditional methods are repetitive, time-consuming, and prone to human error, particularly when handling large volumes of components.

The proposed automated system significantly reduces repetitive manual tasks by introducing a smart measurement and sorting mechanism. The integration of a Wheatstone bridge measurement circuit, ESP32 microcontroller, and servo-based mechanical sorting system eliminates the need for continuous manual identification and storage of resistors.

By automating routine tasks, the system allows students, technicians, and engineers to focus on higher-level activities such as circuit design, analysis, and innovation. This leads to better time utilization and improved productivity in laboratories and workshops.

The project also contributes to workplace safety and ergonomics. Manual reading of resistor color codes for extended periods can cause eye strain and fatigue. Automation minimizes repetitive hand movements and reduces physical strain, thereby supporting healthier working conditions.

Furthermore, the project enhances student skills in embedded systems, automation, robotics, and control engineering. Exposure to such technologies encourages innovation, improves employability, and supports long-term economic growth. Thus, the system strongly supports the objectives of SDG 8.

SDG 9: Industry, Innovation and Infrastructure

SDG 9 focuses on building resilient infrastructure, promoting inclusive industrialization, and fostering innovation. Automation is a key driver of modern industry, and the transition from manual processes to smart systems is an essential step toward Industry 4.0.

The Automated Resistor Sorter and Retriever introduces automation into electronic component management, which is traditionally a manual process in many educational laboratories and small-scale industries. By combining sensors, microcontrollers, and electromechanical actuators, the system demonstrates the practical implementation of smart infrastructure in the electronics domain.

The project serves as a low-cost prototype of automated inventory management. It demonstrates how intelligent systems can improve accuracy, speed, and reliability in component handling. This reduces dependence on manual labour and enhances operational efficiency.

Additionally, the project encourages innovation by integrating hardware design, embedded programming, mechanical systems, and control logic into a single platform. Such interdisciplinary development promotes research and technological advancement in automation and robotics.

The affordability and scalability of the system make it suitable for educational institutions, research laboratories, and small electronics manufacturing units. Therefore, the project contributes to the development of modern and efficient technological infrastructure, directly supporting SDG 9.

SDG 12: Responsible Consumption and Production

SDG 12 emphasizes sustainable resource management and reduction of waste. Electronic waste is a growing global concern, and inefficient handling of electronic components contributes significantly to this problem.

In many laboratories, resistors are frequently misplaced, incorrectly identified, or discarded due to confusion in color code interpretation. As a result, usable components are often wasted, leading to unnecessary financial loss and increased electronic waste.

The proposed system promotes responsible consumption by ensuring accurate identification, organized storage, and easy retrieval of resistors. By maintaining a systematic inventory of components, the likelihood of losing or discarding usable resistors is greatly reduced.

The ability to reuse components efficiently decreases the demand for frequent procurement of new resistors, thereby conserving resources and reducing waste generation. This supports sustainable engineering practices and environmentally responsible laboratory management.

Overall, the Automated Resistor Sorter and Retriever contributes to efficient resource utilization and promotes sustainable production practices in the field of electronics, aligning strongly with SDG 12.

CHAPTER 9

DISCUSSIONS

9.1 Advantages

- **9.1.1 Improves efficiency in sorting electronic components**

The proposed Resistor Sorter and Retriever System significantly improves the efficiency of sorting electronic components by automating the classification process. Manual sorting of resistors based on color codes is time-consuming and prone to human errors, especially when dealing with large quantities. The system utilizes electrical measurement through a Wheatstone bridge and high-resolution ADC (ADS1115) to quickly determine resistor values and sort them accordingly. This reduces manual effort and increases productivity in laboratories and electronic workshops.

- **9.1.2 Provides high accuracy in resistor value detection**

The use of a Wheatstone bridge circuit combined with the ADS1115 analog-to-digital converter ensures precise measurement of resistance values. Unlike traditional methods that rely on visual identification of color bands, the proposed system performs electrical measurement, eliminating errors caused by poor lighting or human misinterpretation. This improves the accuracy and reliability of the sorting process.

- **9.1.3 Enables automated sorting and retrieval**

The system is capable of both sorting and retrieving resistors, making it highly efficient and user-friendly. The ESP32 microcontroller processes the measured resistance values and controls the NEMA 17 stepper motor through the TB6600 driver to position the appropriate storage bin. Servo motors are then used to guide the resistor into the correct compartment. Additionally, the retrieval feature allows users to easily access required resistors, reducing search time.

- **9.1.4 Ensures precise positioning using stepper motor control**

The NEMA 17 stepper motor provides accurate and controlled rotation, enabling precise positioning of the sorting mechanism. The TB6600 driver ensures stable motor operation with proper current control. The inclusion of a homing sensor

(A3144 Hall-effect sensor) allows the system to establish a reference position, ensuring repeatability and minimizing positional errors during operation.

- **9.1.5 Reduces human error and improves reliability**

Manual sorting is highly dependent on human attention and accuracy, which can lead to mistakes. The proposed system eliminates such issues by automating the entire process. The ESP32 executes predefined logic consistently, ensuring reliable and repeatable performance without fatigue or misinterpretation.

- **9.1.6 Compact and cost-effective solution**

The system is built using commonly available components such as ESP32, stepper motors, ADC modules, and servo motors, making it cost-effective compared to industrial sorting machines. Its compact design allows easy integration into small workspaces such as laboratories, educational institutions, and hobby setups.

9.2 Disadvantages

- **9.2.1 Initial setup and calibration complexity**

The system requires proper calibration of the Wheatstone bridge and ADC to ensure accurate resistance measurement. Incorrect calibration may lead to inaccurate sorting. Additionally, setting up motors, sensors, and control logic requires technical expertise, which may be challenging for beginners.

- **9.2.2 Mechanical wear and maintenance**

Since the system uses moving parts such as stepper motors and servo motors, mechanical wear and tear may occur over time. Regular maintenance is required to ensure smooth operation and long-term reliability.

9.3 Applications

- **9.3.1 Electronics laboratories and educational institutions**

The system can be widely used in educational laboratories where students frequently handle resistors. It helps in organizing components efficiently and reduces time spent on manual sorting, thereby improving learning efficiency.

- **9.3.2 Electronic manufacturing and assembly units**

In small-scale manufacturing units, the system can assist in sorting resistors for assembly processes. This improves workflow efficiency and reduces manual labor in component handling.

- **9.3.3 Repair and maintenance workshops**

Technicians often require quick access to resistors of specific values. The retrieval feature of the system allows easy access, saving time during repair and troubleshooting tasks.

- **9.3.4 Inventory management systems**

The system can be integrated into inventory management setups to automatically organize and store resistors based on their values. This ensures better organization and reduces component loss.

- **9.3.5 Hobbyist and DIY electronics projects**

For electronics enthusiasts and hobbyists, the system provides an efficient way to manage and store resistors, making project development more organized and convenient.

9.4 Future Scope

The proposed Resistor Sorter and Retriever System can be further enhanced by incorporating advanced technologies to improve its performance, accuracy, and usability. One of the major improvements could be the integration of IoT capabilities using the ESP32's built-in Wi-Fi functionality. This would allow users to monitor inventory, track resistor usage, and control the system remotely through a mobile application or cloud platform.

Another significant enhancement could be the inclusion of machine vision techniques for resistor identification based on color codes. This would eliminate the need for electrical measurement and further increase speed and flexibility. Additionally, integrating multiple sensing techniques can improve accuracy and enable detection of a wider range of components.

The system can also be upgraded by adding a graphical user interface (GUI) or display module to provide better user interaction and real-time feedback. Voice or app-based control could further enhance usability.

Future versions may include support for sorting other electronic components such as capacitors, diodes, and inductors, making it a universal component sorting system. Improvements in mechanical design, such as faster motors and optimized sorting mechanisms, can further increase efficiency.

Furthermore, implementing advanced algorithms for error detection, auto-calibration, and adaptive sorting can significantly improve system performance. Integration with robotic arms or automated assembly systems can extend its application to industrial automation.

Overall, the system has strong potential for future development into a fully automated and intelligent component management system suitable for educational, industrial, and commercial applications.

CHAPTER 10

CONCLUSION

In conclusion, the proposed Automated Resistor Sorter and Retriever successfully addresses the challenges associated with manual resistor identification, sorting, and retrieval in electronics laboratories and workshops. Resistors are among the most widely used passive components in electronic circuit design, and efficient management of these components is essential for improving productivity, accuracy, and workflow efficiency. Traditional manual sorting methods are time-consuming, prone to human error, and often lead to misplacement or loss of components. The developed system provides an effective automated solution to overcome these limitations.

The system integrates resistance measurement, embedded processing, and electromechanical actuation to achieve a complete automation cycle. By utilizing a Wheatstone bridge-based measurement circuit, the system accurately determines the resistance value of the inserted resistor. The ESP32 microcontroller processes the measured data and controls the sorting and retrieval mechanism using servo motors and a stepper motor, ensuring precise positioning and reliable operation.

The inclusion of an OLED display and user input buttons enhances the interaction between the user and the system by providing real-time display of resistance values and system status. The automated sorting mechanism significantly reduces the time required to organize resistors and eliminates the need for continuous manual identification using color codes or multimeters.

The developed prototype demonstrates improved efficiency, accuracy, and reliability in component handling. By reducing repetitive manual tasks, the system allows students, technicians, and engineers to focus on more productive and creative activities such as circuit design and experimentation. Additionally, the organized storage and retrieval of components minimizes component loss and promotes better resource management in laboratories.

The overall system is designed to be cost-effective, scalable, and suitable for integration into educational institutions, research laboratories, and small-scale electronic manufacturing environments. The project successfully demonstrates the practical implementation of automation, embedded systems, and mechatronics in solving a real-world laboratory problem.

Future enhancements may include the integration of IoT-based inventory management, database storage for component tracking, mobile or web interfaces for remote monitoring, and expansion of storage capacity to support a larger variety of components. These improvements can further increase the efficiency and commercial viability of the system.

In summary, the Automated Resistor Sorter and Retriever provides a reliable and innovative solution for efficient electronic component management and represents a significant step toward smart laboratory automation.

REFERENCES

- [1] J. Sobti, K. K. Dixit, A. Alkhayyat, S. Pant, H. Kaur, and R. Anand, “An Intelligent Robot Sorting System by Deep Learning on RGB-D Image,” in Proc. IEEE Conference, 2024.
- [2] A. Kumar, A. Bapuram, M. A. Devi, Y. Banoth, S. K. Dwivedi, and A. Kumar, “Research on Automatic Sorting System Based on Machine Vision,” in Proc. IEEE Conference, 2024.
- [3] K. D. Klinich, M. A. Manary, N. R. Orton, K. J. Boyle, and J. Hu, “Automated Sorting and Shredding System using Arduino Microcontrollers,” in Proc. IEEE Conference, 2025.
- [4] M. Tonkin, M. C.-P. Kim, and J.-H. Ahn, “Experimental Verification of Automatic Sorting Machine using Infrared Proximity Sensor and Microcontroller,” in Proc. IEEE Conference, 2025.
- [5] A. H. Jalilvand, Z. Kordrostami, and A. Mirzaei, “Sorting It Out in Hardware: A State-of-the-Art Survey,” arXiv preprint, 2023.

APPENDICES

APPENDIX 1:BASE PAPER

Experimental Verification of Automatic Sorting Machine Using Infrared Proximity Sensor and Microcontroller

Darakorn Thappho
Faculty of Engineering
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
60601053@kmitl.ac.th

Sirichai Tammaruckwattana
Faculty of Engineering
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
sirichai.ta@kmitl.ac.th

Abstract—This article proposes the design and builds of an automatic marigold (economic crop) sorting machine by sorting sizing 5 sizes with working together between conveyor belts, electric motors, and microcontroller control unit. The system designed to connect to WIFI to store data on Cloud computing and can display the information through the screen in Real-Time in numeric data format and graphs, Test the accuracy of the system by comparing experiments between microcontroller control unit and PLC control unit, which sold in the market at a high prices but this designed system has a low price and equivalent performance. The results from repeated tests showed that the automatic marigold sorting machine can operate as specified according to the design by separating the size of the marigolds.

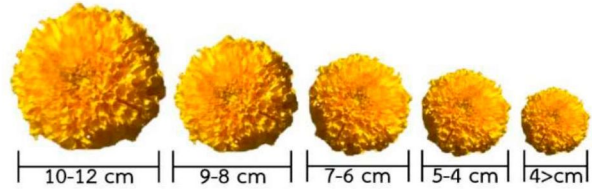
Keywords—automatic, sorting machine, proximity sensor, microcontroller

I. INTRODUCTION

Marigold is an economic plant. There are high market demand and price is stable, can be sold throughout the year and it is a short-lived plant, takes 60 days can picking marigold flowers for sale and can be perennial, flowering for approximately 1 month. Marigold is an economic plant, therefore can create a career for people in the community[1]. For an area of 1 Rai, 5,000 marigolds can be planted. Takes time to care for marigolds plant since the beginning of planting takes 60 days and then from day 61- 120 can be cut marigold flowers other 15-times and 30,000 marigolds can be harvested in each time. There are sizing less than 4 cm, 4-5 cm, 6-7 cm, 8- 9 cm, and 10-12 cm, which the varying prices as shown in Table I. Causing marigold growers to have an income every 3-4 days, While 1 Rai can harvest the produce 15 times. The total duration of planting 1 Rai of marigold flowers is 120 days and the farmers will earn 30,000 Baht/Rai. In 1 year can harvest 3 times, therefore the area of 1 Rai able to generate income up to 120,000 Baht/Rai [2]

TABLE I. TABLE TYPE STYLES

Number	Size (cm)	Price (Bath)
1	10-12	1.00
2	8-9	0.50
3	6-7	0.25
4	4-5	0.10
5	4>	0



From the study, it was found that the number of marigolds collected by farmers each day was high. Size screening is necessary before selling it to the market. For this reason, the researcher has the idea of studying an automatic marigold sorting machine by relying on the motor to drive a conveyor when the object moves to the object detection point and the sensor unit will send signals to the PLC [3,4] control device to processing and after that, the PLC will be the transmitter ordered the air cylinder to work by pushing the object down the specified slot, but the PLC control unit has a quite high cost, therefore, there is a low-cost control unit design that can be used as practically equivalent to the PLC and farmers can use it on their farms at an affordable price, there is also a storage data system for verification so it is suitable and convenient for users by bringing the LOT system and Cloud Computing System for monitoring and processing the number of workpieces on a computer, mobile and tablet quickly and can also store data throughout the year for the farmers enable to analyze the data to compare the products in each season as well and also able to determine the stock of the product.

II. THEORY AND WORK PRINCIPLES

A. Propulsion system

For propulsion, by the conveyor and pulley to transport the objects from the source to the destination, the conveyor belt is commonly used such as flat belt, conveyor-belt and the v- belt conveyor, etc. This experiment uses 2 belt types which are:-

- Conveyor-belt, the material is a PVC green color, waterproof, thickness 3 mm, a width 200 mm, a total length of 5.8 m. used for supporting objects to be size sorting, a medium level of durability, able to support the moderate weight.
- V-Belt conveyor [5,6] the belt helps to transfer power from pulleys to the roller to drive the conveyor using the V-Belt, A-type.

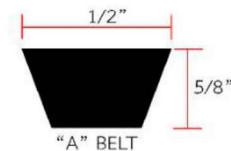


Fig. 1. V-Belt conveyor



Fig. 2. Size of belt A-type

Pulley is an important part of the belt conveyor system that acts in changing directions and control the tension or slack of the conveyor system. As for the motor used as an AC Motor, the rotation of the motor causes the belt to move forward to transport the object from the source to the destination.

The study of devices that will help to control the speed of the motor, Therefore, AC Motor Speed Controller is the equipment chosen for the increase and decrease the voltage to get the required speed in which the resistors can be adjustable.



Fig. 3. Pulley and AC motor

B. Pneumatic

The pneumatic system is a power transmission system that uses air pressure to transmit power by devices such as cylinders that convert wind energy into mechanical energy. The advantage of the pneumatic system is the speed of work. Compressed air has a high operating speed, resistant to explosion because the wind is not combustible and does not explode, therefore it is safe. The equipment in the pneumatic system does not damage when used overpower. It can adjust the speed easily and conveniently to installation. The transfer of air can be easily transmitted through pipes or wind tubes over long distances. Used air can be released into the atmosphere immediately without causing pollution. The air cylinders used in this experiment are two-way cylinders by air using a solenoid valve size 5/2 to control the wind direction.

C. Detection systems or sensors

For the selection of sensors, many factors are to be considered such as the purpose of the work. The worthiness of sensor, controller and sensor compatibility [7] therefore, in this work, choose to use Infrared Proximity Sensor. This sensor is a Diffuse Mode or directly reflects the object by using the principle of refraction of light and reflection with Emitter or Led Infrared as the signal to the receiver or the phototransistor is the receiver to the PLL circuit or Phase Lock Loop. Is an interference filter circuit or disturbance signal and sends it to the output. The object detection only occurs if the object moves nearby the sensor unit which has the spread of the electromagnetic field which generated by the RC oscillator circuit and when the capacitance changes to the one value of capacity will cause a condition called RC resonance which results in the occurrence oscillation of the signal as shown in Fig. 4.

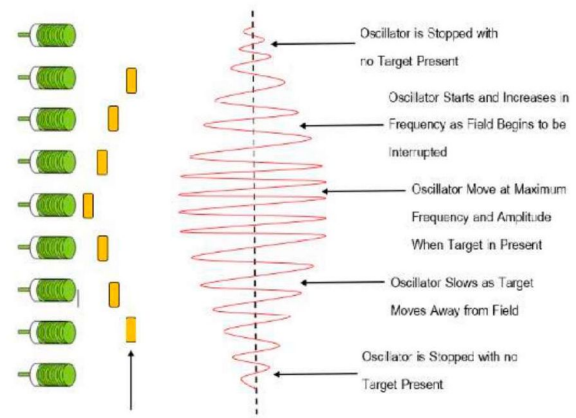


Fig. 4. Range of oscillators while detecting objects of the Proximity sensor

III. THE EXPERIMENTAL

Fig. 5. shows an overview of a control system created to sort the size of marigolds. The conveyor will be driven by AC Motor and take the object along the belt to the detection point where the Infrared Proximity Sensor has installed with a total of 4 stations, each station has been set the height of the detection level at 4 levels from the object release point is 10 cm, 8 cm, 6 cm, and 4 cm. for the last level no sensor installed because the object that can reach to the end is a non-standard sizes that is less than 4 cm in size. For the electricity source that is converted from 220 Volt to direct current 24 Volt to supply to the solenoid valve. For the motor, it has installed devices that will help in controlling the motor speed, it uses an adjustable resistor to increase and decrease the voltage before sending it to AC Motor.

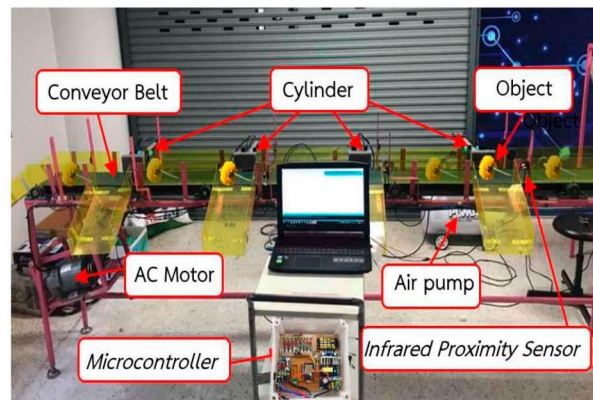


Fig. 5. Automatic size sorter

In the control box section, there are 4 main parts. The first part is the ESP32 board which is used to receive the signal from the sensor and forward it. The second part is the OPTO board which is the signal amplifier used to receive the signal from ESP32 sent to the valve to continue working. The third part is the part of the motor speed controller for adjusting the speed of the belt. The last part is the power supply used to supply power to support the system to operate the control box as shown in Fig. 6.

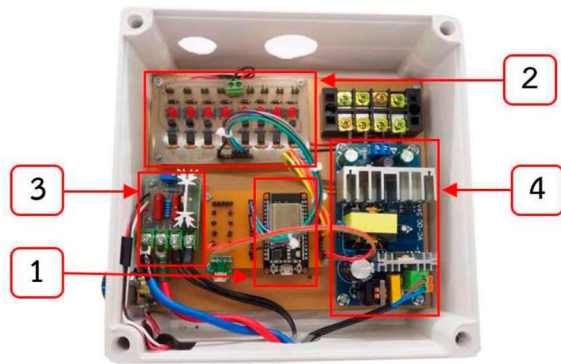


Fig. 6. Microcontroller control unit

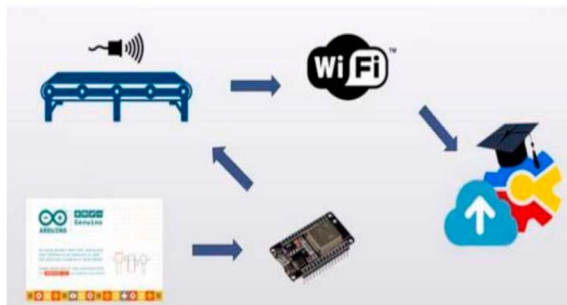


Fig. 7. Flowchart of the sorting machine

IV. STUDY RESULT

Results from the study showed that the size sorting 4 sizes of marigold and 25 flowers of each size by using an Infrared Proximity Sensor which uses multi-size detection of the ESP32 control unit can be used equivalent to a PLC control unit by showing the total workpiece table as shown in Table II and Table III.

TABLE II. RESULT OF CONTROL UNITS

PLC control				
Time (min)	Size (cm)			
	Size A 10-12	Size B 8-9	Size C 6-7	Size D 4-5
5	8	3	7	4
10	13	10	16	12
15	19	17	21	17
20	25	25	25	25

TABLE III. RESULT OF THE EPS32 CONTROL UNITS

EPS32 control				
Time (min)	Size (cm)			
	Size A 10-12	Size B 8-9	Size C 6-7	Size D 4-5
5	6	8	7	8
10	14	13	15	14
15	20	19	21	19
20	25	25	25	25

From the table shows that in each period of 5-20 minutes, different sizes of marigolds were detected between the PLC control unit and the EPS32 control unit. Due to the size of the marigolds tested were mixed in size without screening which

is not the same as the testing of the 2 control sets. However, the test results at the last minute, the automatic.



Fig. 8. Graph of experimental results of PLC control units



Fig. 9. Graph of experimental results of the ESP32 control unit



Fig. 10. The picture shows the sorting of marigold flowers.

Comparison of experimental result graphs between the PLC control unit and ESP32 control unit shown in each period of the size sorting of marigolds in 4 sizes and 25

flowers of each size. At the last moment, the marigold sizes can be sorted with the same results. Therefore, it can be demonstrated that the EPS32 control unit can function as a PLC control unit at a lower cost[8,9,10].

V. SUMMARY OF RESULTS

- Objects to be tested able to sort the size successfully, Sensors for sorting, conveyor belt, and other devices can work correctly according to the model placed.
- From the test of the size sorting of objects, it is found that the ESP32 control unit can be used equivalent to the PLC control unit.
- The part of programming (Software) has been experimented with separating the size of the object by the real machine and found that the size of the object can be separated as needed, also can connect to WIFI to store data to the Cloud and can display the information on the screen in Real-Time both in numeric data format and graphs as well.

REFERENCES

- [1] A. Kanacharoenpong, A. Shutsirung, S. Teplikitkul, J. Konjai and Somporn Choonluchanon, Effect of mixing Media on Growth and Flower Yield of Marigold, *Journal of Agriculture* 19 (2) (2003) 153-159.
- [2] Cowen, R.K.D. and L.C. Ewart. 1990. Inheritance of a male sterile apetalous inflorescence in *Zinnia elegans*. *Acta Horticulturae* 272: 31-40.
- [3] J. Li, L.Y Xiong, N. Peng, B. Dong, P. Wang, L.Q. Liu, Measurement and Control system for Cryogenic Helium Gas Bearing Turbo-expander Experimental Platform Based on Siemens PLC S7-300, in: *Advances in Cryogenic Engineering, 2013 Cryogenic Engineering Conference (CEC) and International Cryogenic Materials Conference (ICMC)*, 20174, p. 1743. Alaska, USA.
- [4] K. Salleh, H. Jasmani, Audit rotation and audit report: empirical evidence from Malaysian PLCs over the period of ten years, *Procedia – Soc. Behav. Sci.* 145 (2014) 40-50.
- [5] Kraisorn Forts and Rich Gold Danai Beach Watcharas, 2556 feeders shrimp automatic rail, *Research University Rajamangala*, 6(1) 63-69.
- [6] M.A. Massoud, M. Sabee, M. Gergais, R. Bakhit, Automated new license plate recognition in Egypt, *Alexandria Eng. J.* 52 (2013) 319-360.
- [7] Wu Wenqi, Xingang Wang, Guan Huang, Xu De, Automatic gear sorting system based on monocular vision, *digital Commun. Networks* 1 (4) (2015) 284-291.
- [8] D.J. Lee, K. James, X. Guangming, Rapid color grading for fruit quality evaluation using direct color mapping, *IEEE Trans. Autom. Sci. Eng.* 8 (2) (2010) 292-303.
- [9] S.-Y. Yang, S.-K. Hsiung, Y.-C. Hung, C.-M. Chang, T.-L. Liao, G.-B. Lee, A cell counting/sorting system incorporated with a microfabricated flow cytometer chip, *Meas. Sci. Technol.* 17 (2006) 2001-2009.
- [10] C.C. Lin, A. Chen, C.H. Lin, Microfluidic cell counter/sorter utilizing multiple particle tracing technique and optically switching approach, *Biomed. Microdevices* 10 (1) (2008) 55-63.

APPENDIX 2:CODE

```

1 #include <Wire.h>
2 #include <Adafruit_ADS1X15.h>
3 #include <ESP32Servo.h>
4 #include <Adafruit_GFX.h>
5 #include <Adafruit_SSD1306.h>
6
7 // ----- OLED -----
8 #define SCREEN_WIDTH 128
9 #define SCREEN_HEIGHT 64
10 #define OLED_ADDR 0x3C
11 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
12
13 // ----- ADC -----
14 Adafruit_ADS1115 ads;
15
16 // ----- SERVOS -----
17 Servo baseServo;
18 Servo topServo;
19 Servo bottomProbe;
20
21 // ----- STEPPER -----
22 #define PULSE_PIN 26
23 #define DIR_PIN 27
24 #define HALL_PIN 33
25
26 const int moveDelay = 1200; // microseconds
27 int currentAngle = 0; // current stepper angle
28
29 // Steps per degree
30 const int stepsPerRev = 200*16; // 16x microstepping
31 const float stepsPerDegree = stepsPerRev / 360.0;
32
33 // ----- BOWLS -----
34 const int bowlAnglesRetrieving[] = {0, 90, 180, 270}; // angles for
    retrieving
35 const int bowlOffsetsSorting[] = {0, 90, 180, 270}; // offsets from
    sorting home
36 const char* resistors[] = {"3.3K", "10K", "100K", "120K"};
37 int indexR = 0;
38
39 // ----- BUTTONS -----
40 #define MODE_BTN 13
41 #define LEFT_BTN 14
42 #define RIGHT_BTN 12
43 #define SELECT_BTN 32
44
45 // ----- SERVOS -----
46 #define BASE_SERVO_PIN 23
47 #define TOP_SERVO_PIN 4
48 #define BOTTOM_SERVO_PIN 5
49
50 // ----- INDICATORS -----
51 #define BUZZER_PIN 17
52 #define BOWL_LED 18
53
54 // ----- BUTTON CONTROL -----
55 unsigned long lastButtonTime = 0;
56 const int buttonDelay = 250;

```

```

57
58 // ----- STATES -----
59 enum State { MODE_SELECT, RETRIEVING, SORTING };
60 State state = MODE_SELECT;
61
62 // ----- BRIDGE CONSTANTS -----
63 float Vin = 3.3;
64 float Rref = 47000.0;
65
66 // ----- RETRIEVING LED -----
67 unsigned long ledStartTime = 0;
68 bool ledActive = false;
69
70 // ----- STARTUP ANIMATION -----
71 #define GEAR_FRAMES 4
72 const unsigned char* gearBitmaps[GEAR_FRAMES] = {
73 /* Add your 4 gear bitmap frames here */
74 };
75
76 // ----- HOMING ANIMATION -----
77 const unsigned char* locBitmaps[2] = {
78 /* Add 2 small location icon frames here */
79 };
80
81 // ----- FUNCTION DECLARATIONS -----
82 void homeMotor();
83 void moveToAngle(int targetAngle);
84 void showMode();
85 void handleModeButton();
86 void showRetrieveUI();
87 void handleRetrieve();
88 void transitionRetrieve(int direction);
89 void handleSorting();
90 void performSorting();
91 void showAnimation(const unsigned char* frames[], int frameCount, int
    duration);
92 void showMessage(String msg);
93 void showMeasuredValue(float Rx);
94
95 // ----- SETUP -----
96 void setup() {
97   Serial.begin(115200);
98   Wire.begin(21,22);
99
100   pinMode(PULSE_PIN, OUTPUT);
101   pinMode(DIR_PIN, OUTPUT);
102   pinMode(HALL_PIN, INPUT_PULLUP);
103
104   pinMode(MODE_BTN, INPUT_PULLUP);
105   pinMode(LEFT_BTN, INPUT_PULLUP);
106   pinMode(RIGHT_BTN, INPUT_PULLUP);
107   pinMode(SELECT_BTN, INPUT_PULLUP);
108
109   pinMode(BUZZER_PIN, OUTPUT);
110   pinMode(BOWL_LED, OUTPUT);
111   digitalWrite(BUZZER_PIN, LOW);
112   digitalWrite(BOWL_LED, LOW);
113

```

```

114 // Servos
115 baseServo.attach(BASE_SERVO_PIN);
116 topServo.attach(TOP_SERVO_PIN);
117 bottomProbe.attach(BOTTOM_SERVO_PIN);
118 baseServo.write(15);
119 topServo.write(90);
120 bottomProbe.write(0);
121
122 // OLED
123 display.begin(SSD1306_SWITCHCAPVCC, OLED_ADDR);
124 display.setTextSize(1);
125 display.setTextColor(SSD1306_WHITE);
126
127 // ADC
128 ads.begin();
129 ads.setGain(GAIN_ONE);
130
131 // ---- Startup animation 2s ----
132 digitalWrite(BUZZER_PIN,HIGH);
133 showAnimation(gearBitmaps, GEAR_FRAMES, 2000); // 2s
134 digitalWrite(BUZZER_PIN,LOW);
135
136 // ---- Credit text 3s ----
137 display.clearDisplay();
138 display.setCursor(0,20);
139 display.setTextSize(1); // normal size to avoid clipping
140 display.println("A product by");
141 display.println("Jubit, Joseph and Noufal");
142 display.display();
143 delay(3000);
144
145 // ---- Homing with location animation ----
146 showMessage("HOMING...");
147 homeMotor(); // run stepper home
148 // Optional: call showAnimation(locBitmaps,2,1000) concurrently during
    homing
149
150 moveToAngle(0); // retrieving home
151 currentAngle = 0;
152
153 showMode();
154 }
155 // ----- LOOP -----
156 void loop() {
157     handleModeButton();
158
159     if(state==RETRIEVING) handleRetrieve();
160     if(state==SORTING) handleSorting();
161
162     // Non-blocking LED for retrieving
163     if(ledActive && millis() - ledStartTime >= 5000){
164         digitalWrite(BOWL_LED,LOW);
165         ledActive = false;
166         showRetrieveUI(); // back to menu
167     }
168 }
169
170 // ----- HOMING -----

```

```

171 void homeMotor() {
172     digitalWrite(DIR_PIN,HIGH);
173     while(digitalRead(HALL_PIN)==HIGH){
174         digitalWrite(PULSE_PIN,HIGH);
175         delayMicroseconds(5000);
176         digitalWrite(PULSE_PIN,LOW);
177         delayMicroseconds(5000);
178     }
179     currentAngle = 0;
180 }
181
182 // ----- MOVE -----
183 void moveToAngle(int targetAngle){
184     int diff = targetAngle - currentAngle;
185     if(diff > 180) diff -= 360;
186     if(diff < -180) diff += 360;
187
188     if(diff >=0) digitalWrite(DIR_PIN,HIGH);
189     else digitalWrite(DIR_PIN,LOW);
190
191     int steps = abs(diff) * stepsPerDegree;
192     for(int i=0;i<steps;i++){
193         digitalWrite(PULSE_PIN,HIGH);
194         delayMicroseconds(moveDelay);
195         digitalWrite(PULSE_PIN,LOW);
196         delayMicroseconds(moveDelay);
197     }
198     currentAngle = targetAngle;
199 }
200
201 // ----- MODE -----
202 void showMode(){
203     display.clearDisplay();
204     display.setCursor(0,10);
205     display.setTextSize(2); // larger for better visibility
206     display.println("Mode Selection");
207     display.setTextSize(1);
208     display.println("-----");
209     display.println("MODE short : Retrieve");
210     display.println("MODE hold  : Sorting");
211     display.display();
212 }
213
214 void handleModeButton(){
215     if(millis()-lastButtonTime < buttonDelay) return;
216     if(digitalRead(MODE_BTN)==LOW){
217         unsigned long t = millis();
218         while(digitalRead(MODE_BTN)==LOW);
219
220         // Beep buzzer on mode change
221         digitalWrite(BUZZER_PIN,HIGH);
222         delay(200);
223         digitalWrite(BUZZER_PIN,LOW);
224
225         if(millis()-t > 1000){
226             // Switch to SORTING
227             state = SORTING;
228             showMessage("SORTING"); // only "Sorting"

```

```

229     moveToAngle(187); // sorting home
230     currentAngle = 187;
231 }else{
232     // Switch to RETRIEVING
233     state = RETRIEVING;
234     moveToAngle(0); // retrieving home
235     currentAngle = 0;
236     showRetrieveUI();
237 }
238 lastButtonTime = millis();
239 }
240 }
241
242 // ----- RETRIEVE -----
243 void showRetrieveUI(){
244     display.clearDisplay();
245     int prev = (indexR-1+4)%4;
246     int next = (indexR+1)%4;
247
248     // Left small text
249     display.setTextSize(1);
250     display.setCursor(0,25);
251     display.println(resistors[prev]);
252
253     // Right small text
254     int textWidth = 6*strlen(resistors[next]);
255     display.setCursor(SCREEN_WIDTH - textWidth,25);
256     display.println(resistors[next]);
257
258     // Center bold
259     display.setTextSize(2);
260     int mainWidth = 12*strlen(resistors[indexR]);
261     display.setCursor(SCREEN_WIDTH/2 - mainWidth/2,25);
262     display.println(resistors[indexR]);
263
264     display.display();
265 }
266
267 void transitionRetrieve(int direction){
268     // Optional smooth sliding animation; implement if desired
269 }
270
271 void handleRetrieve(){
272     if(millis()-lastButtonTime<buttonDelay) return;
273
274     if(digitalRead(LEFT_BTN)==LOW){
275         indexR = (indexR-1+4)%4;
276         showRetrieveUI();
277         lastButtonTime=millis();
278     }
279     if(digitalRead(RIGHT_BTN)==LOW){
280         indexR = (indexR+1)%4;
281         showRetrieveUI();
282         lastButtonTime=millis();
283     }
284
285     if(digitalRead(SELECT_BTN)==LOW && !ledActive){
286         int targetAngle = bowlAnglesRetrieving[indexR];

```

```

287     moveToAngle(targetAngle); // shortest path
288
289     digitalWrite(BOWL_LED,HIGH);
290     ledStartTime = millis();
291     ledActive = true;
292
293     display.clearDisplay();
294     display.setCursor(10,25);
295     display.setTextSize(1);
296     display.println("Take resistor!");
297     display.display();
298
299     lastButtonTime=millis();
300 }
301 }
302
303 // ----- SORTING -----
304 void handleSorting(){
305     if(millis()-lastButtonTime<buttonDelay) return;
306     if(digitalRead(SELECT_BTN)==LOW){
307         performSorting();
308         lastButtonTime=millis();
309     }
310 }
311
312 void performSorting(){
313     topServo.write(0);
314     bottomProbe.write(90);
315     delay(2000);
316
317     long total=0;
318     for(int i=1;i<49;i++){
319         total += ads.readADC_SingleEnded(1);
320         delay(5);
321     }
322     float adc = total/48.0;
323     float V1 = adc * 0.000125;
324     float Rx = (V1*Rref)/(Vin-V1);
325
326     topServo.write(90);
327     bottomProbe.write(0);
328     delay(1000);
329     // Show measured value in large text
330     showMeasuredValue(Rx);
331
332     int bowlIndex;
333     if(Rx < 5500) bowlIndex = 0;
334     else if(Rx < 50000) bowlIndex = 1;
335     else if(Rx < 105000) bowlIndex = 2;
336     else bowlIndex = 3;
337
338     int targetAngle = (187 + bowlOffsetsSorting[bowlIndex]) % 360;
339     moveToAngle(targetAngle);
340
341     baseServo.write(90);
342     delay(2000);
343     baseServo.write(15);
344 }

```

```

345
346 void showMeasuredValue(float Rx){
347     display.clearDisplay();
348     display.setTextSize(2);
349     display.setCursor(20,20);
350     display.print(Rx,0); // show integer value
351     display.println(" Ohm");
352     display.display();
353     delay(2000);
354
355     // Show "Next resistor" in small text
356     display.clearDisplay();
357     display.setTextSize(1);
358     display.setCursor(10,25);
359     display.println("Next resistor");
360     display.display();
361     delay(1500);
362 }
363
364 // ----- ANIMATION -----
365 void showAnimation(const unsigned char* frames[], int frameCount, int
    duration){
366     int frameTime = duration / frameCount;
367     for(int i=0;i<frameCount;i++){
368         display.clearDisplay();
369         // display.drawBitmap(0,0,frames[i], SCREEN_WIDTH, SCREEN_HEIGHT,
            SSD1306_WHITE);
370         display.display();
371         delay(frameTime);
372     }
373 }
374
375 // ----- MESSAGE -----
376 void showMessage(String msg){
377     display.clearDisplay();
378     display.setTextSize(2); // larger messages
379     display.setCursor(10,25);
380     display.println(msg);
381     display.display();
382 }

```

APPENDIX 3:DATA SHEETS

1. Overview

ESP32-WROOM-32 is a powerful, generic Wi-Fi+BT+BLE MCU module that targets a wide variety of applications, ranging from low-power sensor networks to the most demanding tasks, such as voice encoding, music streaming and MP3 decoding.

At the core of this module is the ESP32-D0WDQ6 chip*. The chip embedded is designed to be scalable and adaptive. There are two CPU cores that can be individually controlled, and the CPU clock frequency is adjustable from 80 MHz to 240 MHz. The user may also power off the CPU and make use of the low-power co-processor to constantly monitor the peripherals for changes or crossing of thresholds. ESP32 integrates a rich set of peripherals, ranging from capacitive touch sensors, Hall sensors, SD card interface, Ethernet, high-speed SPI, UART, I²S and I²C.

Note:

* For details on the part numbers of the ESP32 family of chips, please refer to the document [ESP32 Datasheet](#).

The integration of Bluetooth, Bluetooth LE and Wi-Fi ensures that a wide range of applications can be targeted, and that the module is all-around: using Wi-Fi allows a large physical range and direct connection to the Internet through a Wi-Fi router, while using Bluetooth allows the user to conveniently connect to the phone or broadcast low energy beacons for its detection. The sleep current of the ESP32 chip is less than 5 μ A, making it suitable for battery powered and wearable electronics applications. The module supports a data rate of up to 150 Mbps, and 20 dBm output power at the antenna to ensure the widest physical range. As such the module does offer industry-leading specifications and the best performance for electronic integration, range, power consumption, and connectivity.

The operating system chosen for ESP32 is freeRTOS with LwIP; TLS 1.2 with hardware acceleration is built in as well. Secure (encrypted) over the air (OTA) upgrade is also supported, so that users can upgrade their products even after their release, at minimum cost and effort.

Table 1 provides the specifications of ESP32-WROOM-32.

Table 1: ESP32-WROOM-32 Specifications

Categories	Items	Specifications
Certification	RF certification	FCC/CE-RED/IC/TELEC/KCC/SRRC/NCC
	Wi-Fi certification	Wi-Fi Alliance
	Bluetooth certification	BQB
	Green certification	RoHS/REACH
Test	Reliability	HTOL/HTSL/uHAST/TCT/ESD
Wi-Fi	Protocols	802.11 b/g/n (802.11n up to 150 Mbps)
		A-MPDU and A-MSDU aggregation and 0.4 μ s guard interval support
	Frequency range	2.4 GHz ~ 2.5 GHz
Bluetooth	Protocols	Bluetooth v4.2 BR/EDR and BLE specification
	Radio	NZIF receiver with -97 dBm sensitivity
		Class-1, class-2 and class-3 transmitter
		AFH

1. Overview

Categories	Items	Specifications
	Audio	CVSD and SBC
Hardware	Module interfaces	SD card, UART, SPI, SDIO, I ² C, LED PWM, Motor PWM, I ² S, IR, pulse counter, GPIO, capacitive touch sensor, ADC, DAC
	On-chip sensor	Hall sensor
	Integrated crystal	40 MHz crystal
	Integrated SPI flash	4 MB
	Operating voltage/Power supply	3.0 V ~ 3.6 V
	Operating current	Average: 80 mA
	Minimum current delivered by power supply	500 mA
	Recommended operating temperature range	-40 °C ~ +85 °C
	Package size	(18.00±0.10) mm × (25.50±0.10) mm × (3.10±0.10) mm
	Moisture sensitivity level (MSL)	Level 3

2. Pin Definitions

2.1 Pin Layout

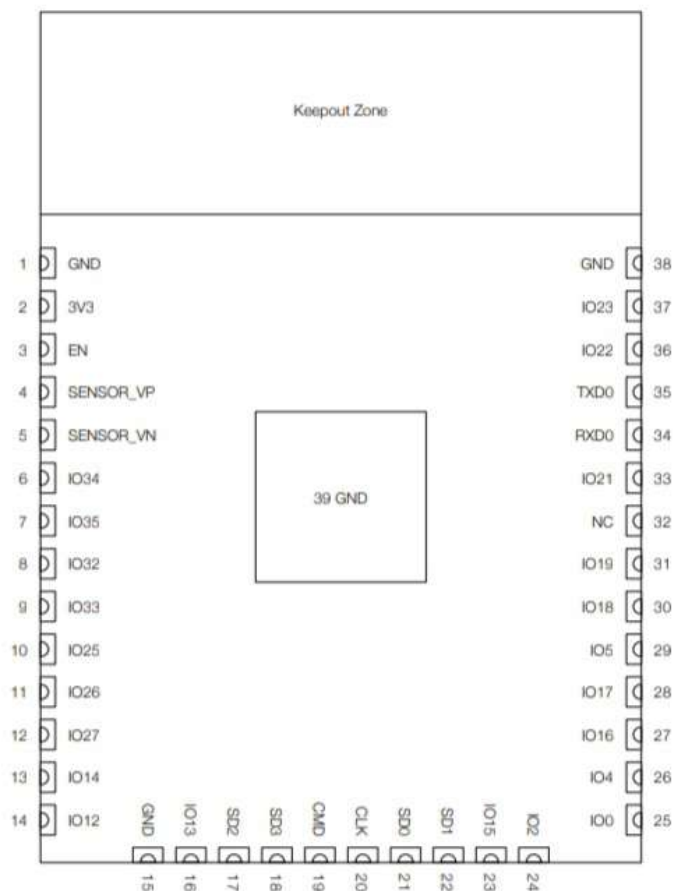


Figure 1: ESP32-WROOM-32 Pin Layout (Top View)

2.2 Pin Description

ESP32-WROOM-32 has 38 pins. See pin definitions in Table 2.

Table 2: Pin Definitions

Name	No.	Type	Function
GND	1	P	Ground
3V3	2	P	Power supply
EN	3	I	Module-enable signal. Active high.
SENSOR_VP	4	I	GPIO36, ADC1_CH0, RTC_GPIO0
SENSOR_VN	5	I	GPIO39, ADC1_CH3, RTC_GPIO3
IO34	6	I	GPIO34, ADC1_CH6, RTC_GPIO4
IO35	7	I	GPIO35, ADC1_CH7, RTC_GPIO5
IO32	8	I/O	GPIO32, XTAL_32K_P (32.768 kHz crystal oscillator input), ADC1_CH4, TOUCH9, RTC_GPIO9
IO33	9	I/O	GPIO33, XTAL_32K_N (32.768 kHz crystal oscillator output), ADC1_CH5, TOUCH8, RTC_GPIO8

2. Pin Definitions

Name	No.	Type	Function
IO25	10	I/O	GPIO25, DAC_1, ADC2_CH8, RTC_GPIO6, EMAC_RXD0
IO26	11	I/O	GPIO26, DAC_2, ADC2_CH9, RTC_GPIO7, EMAC_RXD1
IO27	12	I/O	GPIO27, ADC2_CH7, TOUCH7, RTC_GPIO17, EMAC_RX_DV
IO14	13	I/O	GPIO14, ADC2_CH6, TOUCH6, RTC_GPIO16, MTMS, HSPICLK, HS2_CLK, SD_CLK, EMAC_TXD2
IO12	14	I/O	GPIO12, ADC2_CH5, TOUCH5, RTC_GPIO15, MTDI, HSPIQ, HS2_DATA2, SD_DATA2, EMAC_TXD3
GND	15	P	Ground
IO13	16	I/O	GPIO13, ADC2_CH4, TOUCH4, RTC_GPIO14, MTCK, HSPID, HS2_DATA3, SD_DATA3, EMAC_RX_ER
SHD/SD2*	17	I/O	GPIO9, SD_DATA2, SPIHD, HS1_DATA2, U1RXD
SWP/SD3*	18	I/O	GPIO10, SD_DATA3, SPIWP, HS1_DATA3, U1TXD
SCS/CMD*	19	I/O	GPIO11, SD_CMD, SPICS0, HS1_CMD, U1RTS
SCK/CLK*	20	I/O	GPIO6, SD_CLK, SPICLK, HS1_CLK, U1CTS
SDO/SD0*	21	I/O	GPIO7, SD_DATA0, SPIQ, HS1_DATA0, U2RTS
SDI/SD1*	22	I/O	GPIO8, SD_DATA1, SPID, HS1_DATA1, U2CTS
IO15	23	I/O	GPIO15, ADC2_CH3, TOUCH3, MTDO, HSPICS0, RTC_GPIO13, HS2_CMD, SD_CMD, EMAC_RXD3
IO2	24	I/O	GPIO2, ADC2_CH2, TOUCH2, RTC_GPIO12, HSPiWP, HS2_DATA0, SD_DATA0
IO0	25	I/O	GPIO0, ADC2_CH1, TOUCH1, RTC_GPIO11, CLK_OUT1, EMAC_TX_CLK
IO4	26	I/O	GPIO4, ADC2_CH0, TOUCH0, RTC_GPIO10, HSPiHD, HS2_DATA1, SD_DATA1, EMAC_TX_ER
IO16	27	I/O	GPIO16, HS1_DATA4, U2RXD, EMAC_CLK_OUT
IO17	28	I/O	GPIO17, HS1_DATA5, U2TXD, EMAC_CLK_OUT_180
IO5	29	I/O	GPIO5, VSPICS0, HS1_DATA6, EMAC_RX_CLK
IO18	30	I/O	GPIO18, VSPICLK, HS1_DATA7
IO19	31	I/O	GPIO19, VSPIQ, U0CTS, EMAC_TXD0
NC	32	-	-
IO21	33	I/O	GPIO21, VSPIHD, EMAC_TX_EN
RXD0	34	I/O	GPIO3, U0RXD, CLK_OUT2
TXD0	35	I/O	GPIO1, U0TXD, CLK_OUT3, EMAC_RXD2
IO22	36	I/O	GPIO22, VSPIWP, U0RTS, EMAC_TXD1
IO23	37	I/O	GPIO23, VSPID, HS1_STROBE
GND	38	P	Ground

Notice:

* Pins SCK/CLK, SDO/SD0, SDI/SD1, SHD/SD2, SWP/SD3 and SCS/CMD, namely, GPIO6 to GPIO11 are connected to the integrated SPI flash integrated on the module and are not recommended for other uses.

5. Electrical Characteristics

5.1 Absolute Maximum Ratings

Stresses beyond the absolute maximum ratings listed in Table 4 below may cause permanent damage to the device. These are stress ratings only, and do not refer to the functional operation of the device that should follow the recommended operating conditions.

Table 4: Absolute Maximum Ratings

Symbol	Parameter	Min	Max	Unit
VDD33	Power supply voltage	-0.3	3.6	V
I_{output}^1	Cumulative IO output current	-	1,100	mA
T_{store}	Storage temperature	-40	150	°C

- The module worked properly after a 24-hour test in ambient temperature at 25 °C, and the IOs in three domains (VDD3P3_RTC, VDD3P3_CPU, VDD_SDIO) output high logic level to ground. Please note that pins occupied by flash and/or PSRAM in the VDD_SDIO power domain were excluded from the test.
- Please see Appendix IO_MUX of [ESP32 Datasheet](#) for IO's power domain.

5.2 Recommended Operating Conditions

Table 5: Recommended Operating Conditions

Symbol	Parameter	Min	Typical	Max	Unit
VDD33	Power supply voltage	3.0	3.3	3.6	V
I_{VDD}	Current delivered by external power supply	0.5	-	-	A
T	Operating temperature	-40	-	85	°C

5.3 DC Characteristics (3.3 V, 25 °C)

Table 6: DC Characteristics (3.3 V, 25 °C)

Symbol	Parameter		Min	Typ	Max	Unit
C_{IN}	Pin capacitance		-	2	-	pF
V_{IH}	High-level input voltage		$0.75 \times VDD^1$	-	$VDD^1 + 0.3$	V
V_{IL}	Low-level input voltage		-0.3	-	$0.25 \times VDD^1$	V
I_{IH}	High-level input current		-	-	50	nA
I_{IL}	Low-level input current		-	-	50	nA
V_{OH}	High-level output voltage		$0.8 \times VDD^1$	-	-	V
V_{OL}	Low-level output voltage		-	-	$0.1 \times VDD^1$	V
I_{OH}	High-level source current ($VDD^1 = 3.3$ V, $V_{OH} \geq 2.64$ V, output drive strength set to the maximum)	VDD3P3_CPU power domain ^{1, 2}	-	40	-	mA
		VDD3P3_RTC power domain ^{1, 2}	-	40	-	mA
		VDD_SDIO power domain ^{1, 3}	-	20	-	mA

Symbol	Parameter	Min	Typ	Max	Unit
I_{OL}	Low-level sink current ($VDD^1 = 3.3$ V, $V_{OL} = 0.495$ V, output drive strength set to the maximum)	-	28	-	mA
R_{PU}	Resistance of internal pull-up resistor	-	45	-	k Ω
R_{PD}	Resistance of internal pull-down resistor	-	45	-	k Ω
V_{IL_nRST}	Low-level input voltage of CHIP_PU to power off the chip	-	-	0.6	V

Stepper Motor NEMA 17

This document describes mechanical and electrical specifications for PBC Linear stepper motors; including standard, hollow, and extended shaft variations.

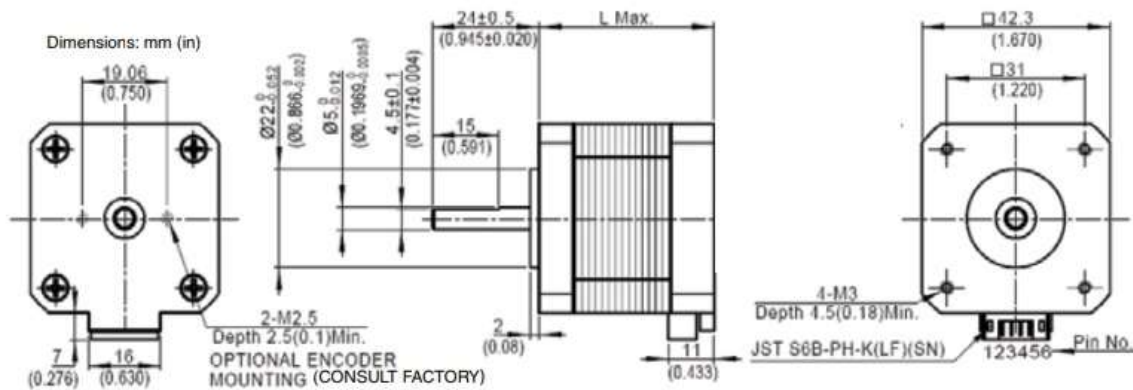


Standard shaft motor shown.

Phases	2
Steps/Revolution	200
Step Accuracy	±5%
Shaft Load	20,000 Hours at 1000 RPM
Axial	25 N (5.6 lbs.) Push 65 N (15 lbs.) Pull
Radial	29 N (6.5 lbs.) At Flat Center
IP Rating	40
Approvals	RoHS
Operating Temp	-20° C to +40° C
Insulation Class	B, 130° C
Insulation Resistance	100 MegOhms

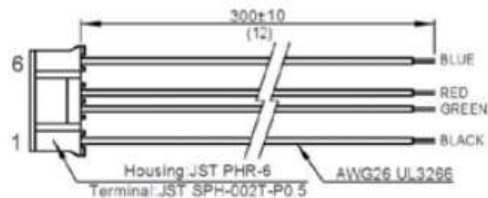
Description	Length	Mounted Rated Current	Mounted Holding Torque	Winding Ohms	mH	Detent Torque	Rotor Inertia	Motor Weight
(Stack)	"L" Max	Amps	Nm Typ. oz-in Typ.	±10% @ 20°C Typ.		mNm oz-in	g cm2 oz-in2	kg lbs
Single	39.8 mm (1.57 in)	2	0.48 68	1.04 2.2		15 2.1	57 0.31	0.28 0.62
Double	48.3 mm (1.90 in)	2	0.63 89	1.3 2.9		25 3.5	82 0.45	0.36 0.79
Triple	62.8 mm (2.47 in)	2	0.83 120	1.49 3.8		30 4.2	123 0.67	0.6 1.3

*All standard motors have plug connector. Consult factory for other options.



Standard shaft dimensions shown. All other dimensions apply to hollow and extended shaft options.

Dimensions: mm (in)
4 Lead Connector, PBC Part#6200490
(Consult factory for optional motor connectors)



+1.815.389.5600 • pbclinear.com



Ball Screws



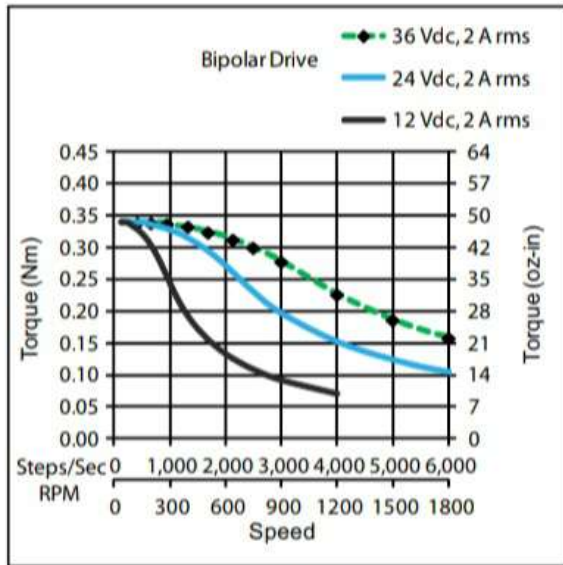
Lead Screws



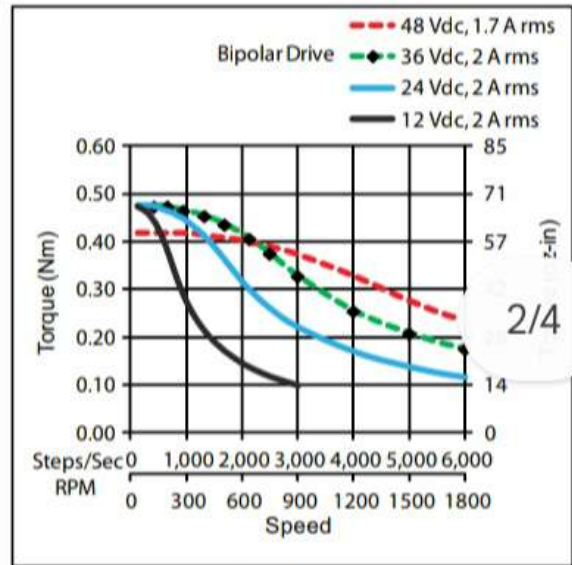
Compact Series

NEMA 17 Stepper Motor

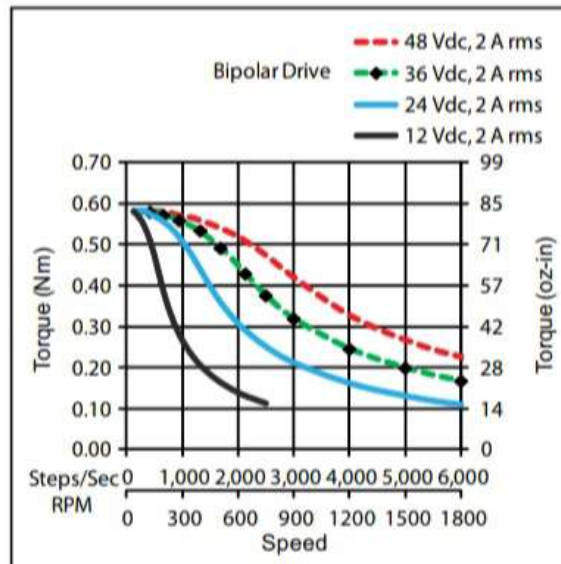
*Performance curves apply to continuous duty cycles.
Consult factory for intermittent cycles or other voltages.



Single Stack



Double Stack



Triple Stack

TB6600

Stepper Motor Driver

Analog Driver Model TB6600

Analog Technology, max. 40 VDC / 4.0 A (PEAK)



Product Description:

The TB6600 single axis drive is a low cost microstepping drive. It is suitable for driving 2-phase and 4-phase hybrid stepper motors. Not for professional applications.

Features:

- Cost-effective
- Supply voltage up to +40 VDC, Output current up to 4.0 A (PEAK)
- Output current selectable in 8 steps via DIP-switch
- Automatic idle-current reduction (in standstill mode) to reduce motor heating
- Pulse input frequency up to 20 kHz
- Input suitable for 5 V signals
- Inputs are optically isolated
- 6 selectable microstep resolutions, up to 6400 steps/rev with standard 1.8° motors
- Suitable for 2-phase and 4-phase motors
- Supports PUL/DIR mode
- Over current and overheat protection

Electrical Specifications:

Parameters	Min	Typ.	Max	Unit
Output current	0.7	-	4.0 (3.5 RMS)	A
Supply voltage	+9	+36	+40	VDC
Logic signal current	8	10	15	mA
Puls input frequency	0	-	20 when duty cycle is 25 high / 75 low 13 when duty cycle is 50 / 50	kHz
Insulation resistance	500			MΩ

Further Specifications:

Microsteps / 1,8 °	200		6400
PUL / DIR		yes	
NEMA sizes	17		24
Motor type Mechatron	42BYGH-XXXX		60BYGH-XXX

22.01.18

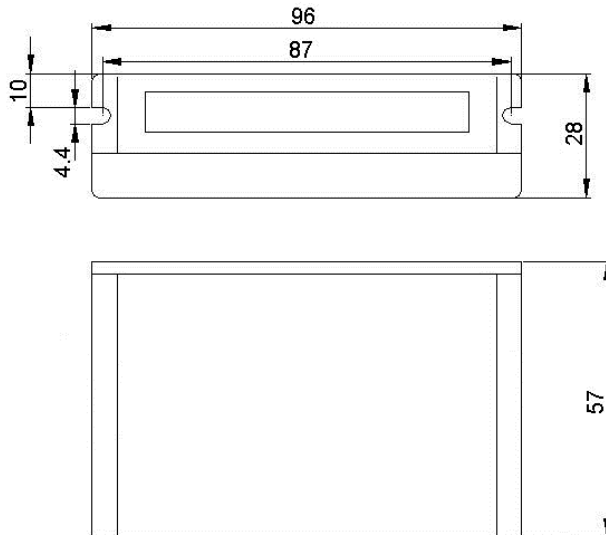
TB6600

Stepper motor driver

Analog Driver

Model TB6600

Mechanical Specifications: (Unit: mm)

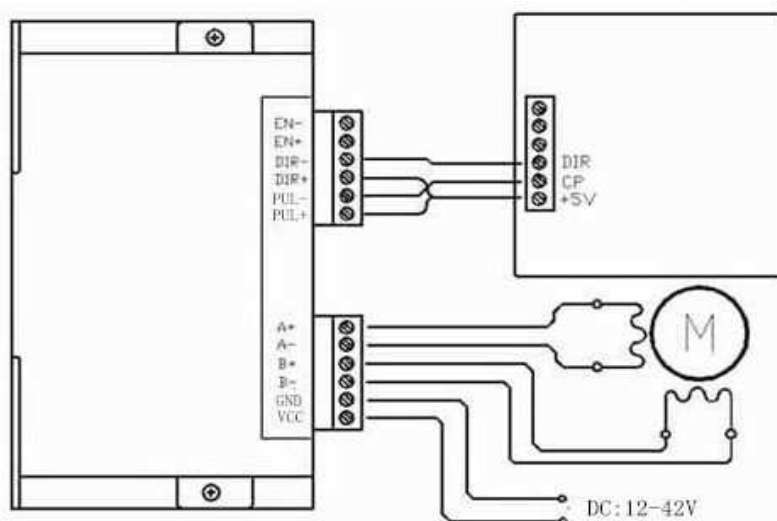


Applications:

Suitable for a wide range of stepping motors of NEMA sizes 17, 23 and 24 (42x42 mm to 60x60 mm). It can be used in various kinds of machines, such as X-Y tables, engraving machines, labeling machines, laser cutters, pick-place devices, and so on. Particularly well suited for applications where low noise levels, less heat development, high speed and high precision are desired.

Typical Connection Schematic:

A typical system consists of stepper motor, stepper motor driver, power supply and controller. The following image shows a typical connection schematic:



Logic control signals which have 5 V can be connected directly;

R 1kΩ must be connected in line when control signal is 12V;

R 2kΩ must be connected in line when control signal is 24V to ensure control signal current is 8mA to 15mA.

22.01.18

Device Comparison Table

DEVICE	RESOLUTION (Bits)	MAXIMUM SAMPLE RATE (SPS)	INPUT CHANNELS Differential (Single-Ended)	PGA	INTERFACE	SPECIAL FEATURES
ADS1115	16	860	2 (4)	Yes	I ² C	Comparator
ADS1114	16	860	1 (1)	Yes	I ² C	Comparator
ADS1113	16	860	1(1)	No	I ² C	None
ADS1015	12	3300	2 (4)	Yes	I ² C	Comparator
ADS1014	12	3300	1 (1)	Yes	I ² C	Comparator
ADS1013	12	3300	1 (1)	No	I ² C	None
ADS1118	16	860	2 (4)	Yes	SPI	Temperature sensor
ADS1018	12	3300	2 (4)	Yes	SPI	Temperature sensor

4 Pin Configuration and Functions

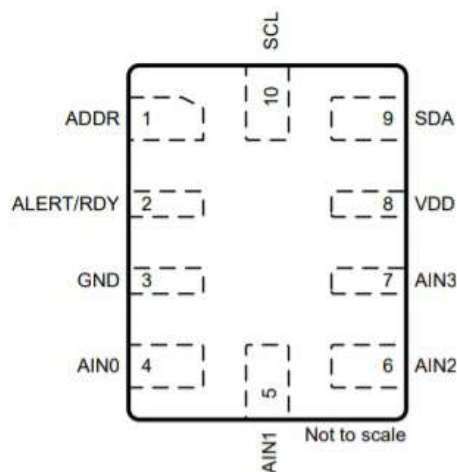


Figure 4-1. RUG Package, 10-Pin (Top View)

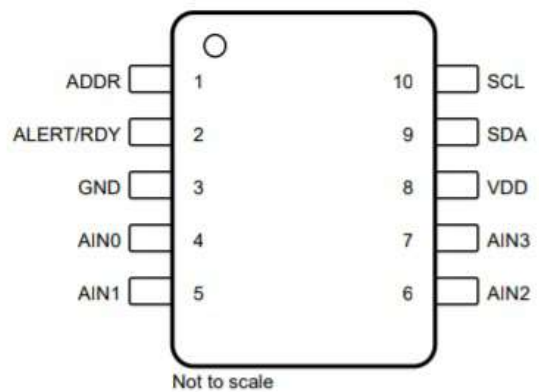


Figure 4-2. DYN and DGS Packages, 10-Pin (Top View)

Table 4-1. Pin Functions: RUG, DYN, and DGS Packages

NAME	PIN			TYPE	DESCRIPTION ⁽¹⁾
	ADS1113	ADS1114	ADS1115		
ADDR	1	1	1	Digital input	I ² C target address select
AIN0	4	4	4	Analog input	Analog input 0
AIN1	5	5	5	Analog input	Analog input 1
AIN2	—	—	6	Analog input	Analog input 2 (ADS1115 only)
AIN3	—	—	7	Analog input	Analog input 3 (ADS1115 only)
ALERT/RDY	—	2	2	Digital output	Comparator output or conversion ready (ADS1114 and ADS1115 only). Open-drain output. Connect to VDD using a pullup resistor.
GND	3	3	3	Analog	Ground
NC	2, 6, 7	6, 7	—	—	No connect. Leave the pin floating or connect to GND.
SCL	10	10	10	Digital input	Serial clock input. Connect to VDD using a pullup resistor.
SDA	9	9	9	Digital I/O	Serial data input and output. Connect to VDD using a pullup resistor.
VDD	8	8	8	Analog	Power supply. Connect a 0.1μF, power-supply decoupling capacitor to GND.

(1) See the [Unused Inputs and Outputs](#) section for unused pin connections.

5 Specifications

5.1 Absolute Maximum Ratings

over operating ambient temperature range (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT
Power-supply voltage	VDD to GND	−0.3	7	V
Analog input voltage	AIN0, AIN1, AIN2, AIN3	GND − 0.3	VDD + 0.3	V
Digital input voltage	SDA, SCL, ADDR, ALERT/RDY	GND − 0.3	5.5	V
Input current, continuous	Any pin except power supply pins	−10	10	mA
Temperature	Operating ambient, T _A	−40	125	°C
	Junction, T _J	−40	150	
	Storage, T _{stg}	−60	150	

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

5.2 ESD Ratings

			VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±2000	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±500	

- (1) JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.
 (2) JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

		MIN	NOM	MAX	UNIT
POWER SUPPLY					
	Power supply (VDD to GND)	2		5.5	V
ANALOG INPUTS⁽¹⁾					
FSR	Full-scale input voltage range ⁽²⁾ (V _{IN} = V _(AINP) − V _(AINN))	±0.256		±6.144	V
V _(AINx)	Absolute input voltage	GND		VDD	V
DIGITAL INPUTS					
V _{DIG}	Digital input voltage	GND		5.5	V
TEMPERATURE					
T _A	Operating ambient temperature	−40		125	°C

- (1) AINP and AINN denote the selected positive and negative inputs. AINx denotes one of the four available analog inputs.
 (2) This parameter expresses the full-scale range of the ADC scaling. No more than VDD + 0.3V must be applied to the analog inputs of the device. See [Table 7-1](#) for more information.

5.4 Thermal Information

THERMAL METRIC ⁽¹⁾		RUG (X2QFN)	DYN (SOT)	DGS (VSSOP)	UNIT
		10 PINS	10 PINS	10 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	245.2	147.1	182.7	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	69.3	59.3	67.2	°C/W
R _{θJB}	Junction-to-board thermal resistance	172.0	71.3	103.8	°C/W
ψ _{JT}	Junction-to-top characterization parameter	8.2	2.8	10.2	°C/W
ψ _{JB}	Junction-to-board characterization parameter	170.8	70.4	102.1	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	N/A	N/A	N/A	°C/W

- (1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application note.

5.5 Electrical Characteristics

at VDD = 3.3V, data rate = 8SPS, and full-scale input voltage range (FSR) = $\pm 2.048\text{V}$ (unless otherwise noted); maximum and minimum specifications apply from T_A = -40°C to $+125^{\circ}\text{C}$; typical specifications are at T_A = 25°C

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
ANALOG INPUT						
	Common-mode input impedance	FSR = ±6.144V ⁽¹⁾		10		MΩ
		FSR = ±4.096V ⁽¹⁾ , FSR = ±2.048V		6		
		FSR = ±1.024V		3		
		FSR = ±0.512V, FSR = ±0.256V		100		
	Differential input impedance	FSR = ±6.144V ⁽¹⁾		22		MΩ
		FSR = ±4.096V ⁽¹⁾		15		
		FSR = ±2.048V		4.9		
		FSR = ±1.024V		2.4		
		FSR = ±0.512V, ±0.256V		710		kΩ
SYSTEM PERFORMANCE						
	Resolution (no missing codes)		16			Bits
DR	Data rate		8, 16, 32, 64, 128, 250, 475, 860			SPS
	Data rate variation	All data rates	–10%		10%	
	Output noise		See Noise Performance section			
INL	Integral nonlinearity	DR = 8SPS, FSR = ±2.048V ⁽²⁾			1	LSB
	Offset error	FSR = ±2.048V, differential inputs	–3	±1	3	LSB
		FSR = ±2.048V, single-ended inputs		±3		
	Offset drift over temperature	FSR = ±2.048V		0.005		LSB/°C
	Long-term Offset drift	FSR = ±2.048V, T _A = 125°C, 1000 hours		±1		LSB
	Offset power-supply rejection	FSR = ±2.048V, DC supply variation		1		LSB/V
	Offset channel match	Match between any two inputs		3		LSB
	Gain error ⁽³⁾	FSR = ±2.048V, T _A = 25°C		0.01%	0.15%	
	Gain drift over temperature ⁽³⁾	FSR = ±0.256V		7		ppm/°C
		FSR = ±2.048V		5	40	
		FSR = ±6.144V ⁽¹⁾		5		
	Long-term gain drift ⁽³⁾	FSR = ±2.048V, T _A = 125°C, 1000 hours		±0.05%		
	Gain power-supply rejection			80		ppm/V
	Gain match ⁽³⁾	Match between any two gains		0.02%	0.1%	
	Gain channel match	Match between any two inputs		0.05%	0.1%	
CMRR	Common-mode rejection ratio	At DC, FSR = ±0.256V		105		dB
		At DC, FSR = ±2.048V		100		
		At DC, FSR = ±6.144V ⁽¹⁾		90		
		f _{CM} = 60Hz, DR = 8SPS		105		
		f _{CM} = 50Hz, DR = 8SPS		105		
DIGITAL INPUT/OUTPUT						
V _{IH}	High-level input voltage		0.7 VDD		5.5	V
V _{IL}	Low-level input voltage		GND		0.3 VDD	V
V _{OL}	Low-level output voltage	I _{OL} = 3mA	GND	0.15	0.4	V
	Input leakage current	GND < V _{DIG} < VDD	–10		10	μA

5.5 Electrical Characteristics (continued)

at VDD = 3.3V, data rate = 8SPS, and full-scale input voltage range (FSR) = ±2.048V (unless otherwise noted); maximum and minimum specifications apply from TA = –40°C to +125°C; typical specifications are at TA = 25°C

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
POWER-SUPPLY					
I _{VDD}	Supply current	Power-down	TA = 25°C	0.5	μA
				5	
		Operating	TA = 25°C	150	
				200	
P _D	Power dissipation	VDD = 5.0V		0.9	mW
		VDD = 3.3 V		0.5	
		VDD = 2.0V		0.3	

- (1) This parameter expresses the full-scale range of the ADC scaling. No more than VDD + 0.3V must be applied to the analog inputs of the device. See Table 7-1 for more information.
- (2) Best-fit INL; covers 99% of full-scale
- (3) Includes all errors from onboard PGA and voltage reference

5.6 Timing Requirements: I²C

over operating ambient temperature range and VDD = 2.0V to 5.5V (unless otherwise noted)

		FAST MODE		HIGH-SPEED MODE		UNIT
		MIN	MAX	MIN	MAX	
f _{SCL}	SCL clock frequency	0.01	0.4	0.01	3.4	MHz
t _{BUF}	Bus free time between START and STOP condition	600		160		ns
t _{HDSTA}	Hold time after repeated START condition. After this period, the first clock is generated.	600		160		ns
t _{SUSTA}	Setup time for a repeated START condition	600		160		ns
t _{SUSTO}	Setup time for STOP condition	600		160		ns
t _{HDDAT}	Data hold time	0		0		ns
t _{SUDAT}	Data setup time	100		10		ns
t _{LOW}	Low period of the SCL clock pin	1300		160		ns
t _{HIGH}	High period for the SCL clock pin	600		60		ns
t _F	Fall time for both SDA and SCL signals ⁽¹⁾		300		160	ns
t _R	Rise time for both SDA and SCL signals ⁽¹⁾		300		160	ns

- (1) For high-speed mode maximum values, the capacitive load on the bus line must not exceed 400pF.

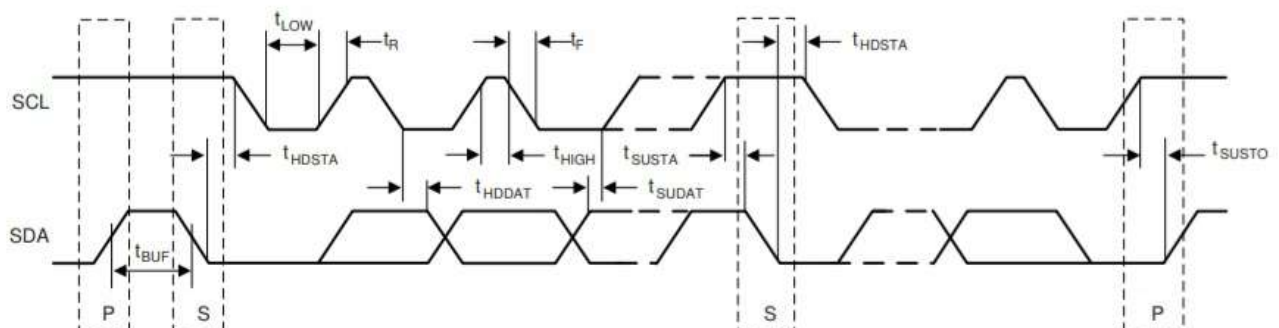


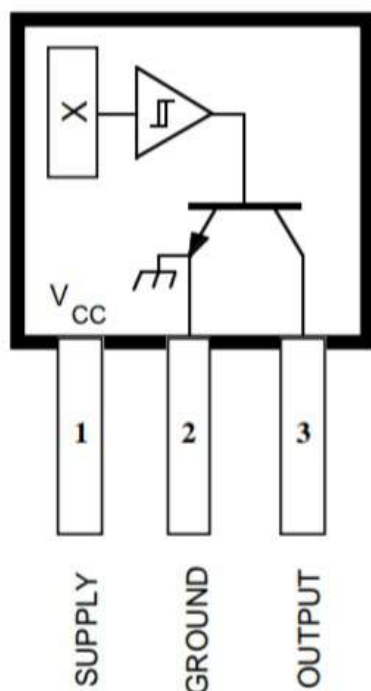
Figure 5-1. I²C Interface Timing

3141 THRU 3144

1/8

Data Sheet
27621.6B*

SENSITIVE HALL-EFFECT SWITCHES FOR HIGH-TEMPERATURE OPERATION



Dwg. PH-003A

Pinning is shown viewed from branded side.

ABSOLUTE MAXIMUM RATINGS at $T_A = +25^\circ\text{C}$

Supply Voltage, V_{CC}	28 V
Reverse Battery Voltage, V_{RCC}	-35 V
Magnetic Flux Density, B	Unlimited
Output OFF Voltage, V_{OUT}	28 V
Reverse Output Voltage, V_{OUT}	-0.5 V
Continuous Output Current, I_{OUT}	25 mA
Operating Temperature Range, T_A	
Suffix 'E-'	-40°C to +85°C
Suffix 'L-'	-40°C to +150°C
Storage Temperature Range, T_S	-65°C to +170°C

These Hall-effect switches are monolithic integrated circuits with tighter magnetic specifications, designed to operate continuously over extended temperatures to +150°C, and are more stable with both temperature and supply voltage changes. The unipolar switching characteristic makes these devices ideal for use with a simple bar or rod magnet. The four basic devices (3141, 3142, 3143, and 3144) are identical except for magnetic switch points.

Each device includes a voltage regulator for operation with supply voltages of 4.5 to 24 volts, reverse battery protection diode, quadratic Hall-voltage generator, temperature compensation circuitry, small-signal amplifier, Schmitt trigger, and an open-collector output to sink up to 25 mA. With suitable output pull up, they can be used with bipolar or CMOS logic circuits. The A3141- and A3142- are improved replacements for the UGN/UGS3140-; the A3144- is the improved replacement for the UGN/UGS3120-.

The first character of the part number suffix determines the device operating temperature range. Suffix 'E-' is for the automotive and industrial temperature range of -40°C to +85°C. Suffix 'L-' is for the automotive and military temperature range of -40°C to +150°C. Three package styles provide a magnetically optimized package for most applications. Suffix '-LT' is a miniature SOT89/TO-243AA transistor package for surface-mount applications; suffix '-UA' is a three-lead ultra-mini-SIP.

FEATURES and BENEFITS

- Superior Temp. Stability for Automotive or Industrial Applications
- 4.5 V to 24 V Operation ... Needs Only An Unregulated Supply
- Open-Collector 25 mA Output ... Compatible with Digital Logic
- Reverse Battery Protection
- Activate with Small, Commercially Available Permanent Magnets
- Solid-State Reliability
- Small Size
- Resistant to Physical Stress

Always order by complete part number, e.g., **A3141ELT**.

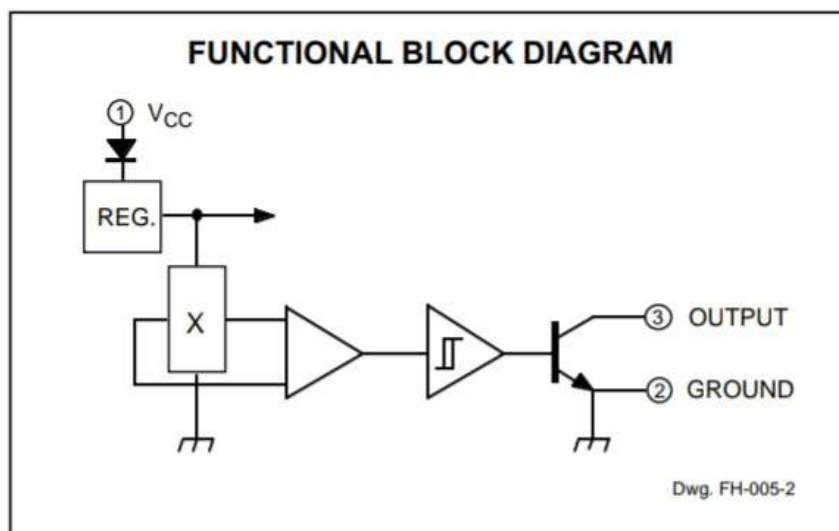


3141 THRU 3144

SENSITIVE

HALL-EFFECT SWITCHES

FOR HIGH-TEMP. OPERATION



ELECTRICAL CHARACTERISTICS at $V_{CC} = 8\text{ V}$ over operating temperature range.

Characteristic	Symbol	Test Conditions	Limits			
			Min.	Typ.	Max.	Units
Supply Voltage	V_{CC}	Operating	4.5	—	24	V
Output Saturation Voltage	$V_{OUT(SAT)}$	$I_{OUT} = 20\text{ mA}$, $B > B_{OP}$	—	175	400	mV
Output Leakage Current	I_{OFF}	$V_{OUT} = 24\text{ V}$, $B < B_{RP}$	—	<1.0	10	μA
Supply Current	I_{CC}	$B < B_{RP}$ (Output OFF)	—	4.4	9.0	mA
Output Rise Time	t_r	$R_L = 820\ \Omega$, $C_L = 20\text{ pF}$	—	0.04	2.0	μs
Output Fall Time	t_f	$R_L = 820\ \Omega$, $C_L = 20\text{ pF}$	—	0.18	2.0	μs

MAGNETIC CHARACTERISTICS in gauss over operating supply voltage range.

Characteristic	Part Numbers*											
	A3141—			A3142—			A3143—			A3144—		
	Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.
B_{OP} at $T_A = 25^\circ\text{C}$	50	100	160	130	180	230	220	280	340	70	—	350
over operating temp. range	30	100	175	115	180	245	205	280	355	35	—	450
B_{RP} at $T_A = 25^\circ\text{C}$	10	45	130	75	125	175	165	225	285	50	—	330
over operating temp. range	10	45	145	60	125	190	150	225	300	25	—	430
B_{hys} at $T_A = 25^\circ\text{C}$	20	55	80	30	55	80	30	55	80	20	55	—
over operating temp. range	20	55	80	30	55	80	30	55	80	20	55	—

NOTES: Typical values are at $T_A = +25^\circ\text{C}$ and $V_{CC} = 8\text{ V}$.

B_{OP} = operate point (output turns ON); B_{RP} = release point (output turns OFF); B_{hys} = hysteresis ($B_{OP} - B_{RP}$).

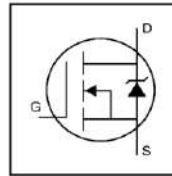
1 gauss (G) is exactly equal to 0.1 millitesla (mT).

*Complete part number includes a suffix to identify operating temperature range (E- or L-) and package type (-LT or -UA).



115 Northeast Cutoff, Box 15036
Worcester, Massachusetts 01615-0036 (508) 853-5000
Copyright © 1993, 2002 Allegro Microsystems, Inc.

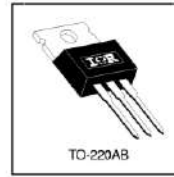
- Logic-Level Gate Drive
- Advanced Process Technology
- Dynamic dv/dt Rating
- 175°C Operating Temperature
- Fast Switching
- Fully Avalanche Rated



$$V_{DS} = 55V$$

$$R_{DS(on)} = 0.022\Omega$$

$$I_D = 47A$$



Description

Fifth Generation HEXFETs from International Rectifier utilize advanced processing techniques to achieve the lowest possible on-resistance per silicon area. This benefit, combined with the fast switching speed and ruggedized device design that HEXFET Power MOSFETs are well known for, provides the designer with an extremely efficient device for use in a wide variety of applications.

The TO-220 package is universally preferred for all commercial-industrial applications at power dissipation levels to approximately 50 watts. The low thermal resistance and low package cost of the TO-220 contribute to its wide acceptance throughout the industry.

Absolute Maximum Ratings

	Parameter	Max.	Units
I_D @ $T_C = 25^\circ\text{C}$	Continuous Drain Current, $V_{GS} = 10V$	47	A
I_D @ $T_C = 100^\circ\text{C}$	Continuous Drain Current, $V_{GS} = 10V$	33	A
I_{DM}	Pulsed Drain Current ①	160	A
P_D @ $T_C = 25^\circ\text{C}$	Power Dissipation	110	W
	Linear Derating Factor	0.71	W/°C
V_{GS}	Gate-to-Source Voltage	± 16	V
E_{AS}	Single Pulse Avalanche Energy ②	210	mJ
I_{AR}	Avalanche Current ③	25	A
E_{ARR}	Repetitive Avalanche Energy ③	11	mJ
dv/dt	Peak Diode Recovery dv/dt ③	5.0	V/ns
T_J	Operating Junction and Storage Temperature Range	-55 to +175	°C
	Soldering Temperature, for 10 seconds	300 (1.6mm from case)	
	Mounting torque, 6-32 or M3 screw	10 lb-in (1.1N-m)	

Thermal Resistance

	Parameter	Min.	Typ.	Max.	Units
R_{JC}	Junction-to-Case	—	—	1.4	°C/W
R_{CS}	Case-to-Sink, Flat, Greased Surface	—	0.50	—	
R_{JA}	Junction-to-Ambient	—	—	62	

8/25/97

IRLZ44N

Electrical Characteristics @ $T_J = 25^\circ\text{C}$ (unless otherwise specified)

Parameter	Min.	Typ.	Max.	Units	Conditions
$V_{(BR)DSS}$	55	—	—	V	$V_{GS} = 0V, I_D = 250\mu A$
$\Delta V_{(BR)DSS}/\Delta T_J$	0.070	—	—	V/°C	Reference to $25^\circ\text{C}, I_D = 1mA$
$R_{DS(on)}$	—	0.022	—	Ω	$V_{GS} = 10V, I_D = 25A$ ④
	—	0.025	—		$V_{GS} = 5.0V, I_D = 25A$ ④
	—	0.035	—		$V_{GS} = 4.0V, I_D = 21A$ ④
$V_{GS(th)}$	1.0	2.0	—	V	$V_{DS} = V_{GS}, I_D = 250\mu A$
g_m	21	—	—	S	$V_{DS} = 25V, I_D = 25A$
I_{DSS}	—	25	—	μA	$V_{DS} = 55V, V_{GS} = 0V$
	—	250	—		$V_{DS} = 44V, V_{GS} = 0V, T_J = 150^\circ\text{C}$
I_{DSS}	—	100	—	nA	$V_{DS} = 16V$
	—	-100	—		$V_{GS} = -16V$
Q_g	—	48	—	nC	$I_D = 25A$
Q_{gs}	—	8.6	—		$V_{DS} = 44V$
Q_{gd}	—	25	—		$V_{DS} = 5.0V$, See Fig. 6 and 13 ④
t_{don}	—	11	—	ns	$V_{DD} = 28V$
t_r	—	84	—		$I_D = 25A$
$t_{d(off)}$	—	26	—		$R_G = 3.4\Omega, V_{GS} = 5.0V$
t_f	—	15	—		$R_D = 1.1\Omega$, See Fig. 10 ④
L_D	4.5	—	—	nH	Between lead, 6mm (0.25in.) from package and center of die contact
L_S	7.5	—	—		
C_{iss}	—	1700	—	pF	$V_{DS} = 0V$
C_{oss}	—	400	—		$V_{DS} = 25V$
C_{rss}	—	150	—		$f = 1\text{ MHz}$, See Fig. 5

Source-Drain Ratings and Characteristics

Parameter	Min.	Typ.	Max.	Units	Conditions
I_S	—	—	47	A	MOSFET symbol showing the integral reverse p-n junction diode.
I_{SM}	—	—	160	A	
V_{SD}	—	—	1.3	V	$T_J = 25^\circ\text{C}, I_S = 25A, V_{GS} = 0V$ ④
t_{rr}	—	80	120	ns	$T_J = 25^\circ\text{C}, I_S = 25A$
Q_{rr}	—	210	320	nC	$dv/dt = 100A/\mu s$ ④
t_{on}	—	—	—	—	Intrinsic turn-on time is negligible (turn-on is dominated by $L_S + L_D$)

Notes:

① Repetitive rating; pulse width limited by max. junction temperature. (See fig. 11)
② $V_{DD} = 25V$, starting $T_J = 25^\circ\text{C}$; $L = 470\mu H$
 $R_G = 25\Omega, I_{AS} = 25A$. (See Figure 12)

③ $I_{SD} \leq 25A, dv/dt \leq 270A/\mu s, V_{DD} \leq V_{(BR)DSS}, T_J \leq 175^\circ\text{C}$
④ Pulse width $\leq 300\mu s$; duty cycle $\leq 2\%$.

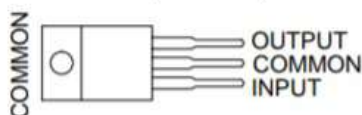
μA7800 SERIES POSITIVE-VOLTAGE REGULATORS

SLVS056J – MAY 1976 – REVISED MAY 2003

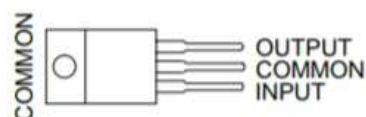
- 3-Terminal Regulators
- Output Current up to 1.5 A
- Internal Thermal-Overload Protection

- High Power-Dissipation Capability
- Internal Short-Circuit Current Limiting
- Output Transistor Safe-Area Compensation

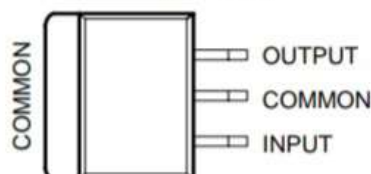
**KC (TO-220) PACKAGE
(TOP VIEW)**



**KCS (TO-220) PACKAGE
(TOP VIEW)**



**KTE PACKAGE
(TOP VIEW)**



description/ordering information

This series of fixed-voltage integrated-circuit voltage regulators is designed for a wide range of applications. These applications include on-card regulation for elimination of noise and distribution problems associated with single-point regulation. Each of these regulators can deliver up to 1.5 A of output current. The internal current-limiting and thermal-shutdown features of these regulators essentially make them immune to overload. In addition to use as fixed-voltage regulators, these devices can be used with external components to obtain adjustable output voltages and currents, and also can be used as the power-pass element in precision regulators.

ORDERING INFORMATION

T _J	V _{O(NOM)} (V)	PACKAGE†		ORDERABLE PART NUMBER	TOP-SIDE MARKING
0°C to 125°C	5	POWER-FLEX (KTE)	Reel of 2000	μA7805CKTER	μA7805C
		TO-220 (KC)	Tube of 50	μA7805CKC	μA7805C
		TO-220, short shoulder (KCS)	Tube of 20	μA7805CKCS	
	8	POWER-FLEX (KTE)	Reel of 2000	μA7808CKTER	μA7808C
		TO-220 (KC)	Tube of 50	μA7808CKC	μA7808C
		TO-220, short shoulder (KCS)	Tube of 20	μA7808CKCS	
	10	POWER-FLEX (KTE)	Reel of 2000	μA7810CKTER	μA7810C
		TO-220 (KC)	Tube of 50	μA7810CKC	μA7810C
	12	POWER-FLEX (KTE)	Reel of 2000	μA7812CKTER	μA7812C
		TO-220 (KC)	Tube of 50	μA7812CKC	μA7812C
		TO-220, short shoulder (KCS)	Tube of 20	μA7812CKCS	
	15	POWER-FLEX (KTE)	Reel of 2000	μA7815CKTER	μA7815C
		TO-220 (KC)	Tube of 50	μA7815CKC	μA7815C
		TO-220, short shoulder (KCS)	Tube of 20	μA7815CKCS	
	24	POWER-FLEX (KTE)	Reel of 2000	μA7824CKTER	μA7824C
		TO-220 (KC)	Tube of 50	μA7824CKC	μA7824C

† Package drawings, standard packing quantities, thermal data, symbolization, and PCB design guidelines are available at www.ti.com/sc/package.



Please be aware that an important notice concerning availability, standard warranty, and use in critical applications of Texas Instruments semiconductor products and disclaimers thereto appears at the end of this data sheet.

PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

**TEXAS
INSTRUMENTS**

POST OFFICE BOX 655303 • DALLAS, TEXAS 75265

Copyright © 2003, Texas Instruments Incorporated

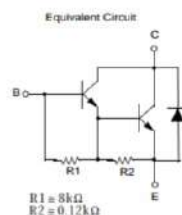
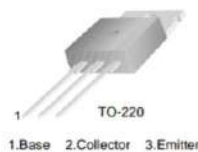


ON Semiconductor®

TIP120 / TIP121 / TIP122 NPN Epitaxial Darlington Transistor

Features

- Medium Power Linear Switching Applications
- Complementary to TIP125 / TIP126 / TIP127



Ordering Information

Part Number	Top Mark	Package	Packing Method
TIP120	TIP120	TO-220 3L (Single Gauge)	Bulk
TIP120TU	TIP120	TO-220 3L (Single Gauge)	Rail
TIP121	TIP121	TO-220 3L (Single Gauge)	Bulk
TIP121TU	TIP121	TO-220 3L (Single Gauge)	Rail
TIP122	TIP122	TO-220 3L (Single Gauge)	Bulk
TIP122TU	TIP122	TO-220 3L (Single Gauge)	Rail

Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at $T_C = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	Value	Unit
V_{CB0}	Collector-Base Voltage	TIP120	60
		TIP121	80
		TIP122	100
V_{CEO}	Collector-Emitter Voltage	TIP120	60
		TIP121	80
		TIP122	100
V_{EBO}	Emitter-Base Voltage	5	V
I_C	Collector Current (DC)	5	A
I_{CP}	Collector Current (Pulse)	8	A
I_B	Base Current (DC)	120	mA
T_J	Junction Temperature	150	$^\circ\text{C}$
T_{STG}	Storage Temperature Range	-65 to 150	$^\circ\text{C}$

© 2001 Semiconductor Components Industries, LLC.
November 2017, Rev.2

Publication Order Number:
TIP122/D

Thermal Characteristics

Values are at $T_C = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	Value	Unit
P_C	Collector Dissipation ($T_A = 25^\circ\text{C}$)	2	W
	Collector Dissipation ($T_C = 25^\circ\text{C}$)	65	

Electrical Characteristics

Values are at $T_C = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	Conditions	Min.	Max.	Unit
$V_{CE0(sus)}$	Collector-Emitter Sustaining Voltage	TIP120	60		V
		TIP121	80		
		TIP122	100		
I_{CEO}	Collector Cut-Off Current	TIP120		0.5	mA
		TIP121		0.5	
		TIP122		0.5	
I_{CBO}	Collector Cut-Off Current	TIP120		0.2	mA
		TIP121		0.2	
		TIP122		0.2	
I_{EBO}	Emitter Cut-Off Current	$V_{EB} = 5\text{ V}, I_C = 0$		2	mA
h_{FE}	DC Current Gain ⁽¹⁾	$V_{CE} = 3\text{ V}, I_C = 0.5\text{ A}$	1000		
		$V_{CE} = 3\text{ V}, I_C = 3\text{ A}$	1000		
$V_{CE(sat)}$	Collector-Emitter Saturation Voltage ⁽¹⁾	$I_C = 3\text{ A}, I_B = 12\text{ mA}$		2.0	V
		$I_C = 5\text{ A}, I_B = 20\text{ mA}$		4.0	
$V_{BE(on)}$	Base-Emitter On Voltage ⁽¹⁾	$V_{CE} = 3\text{ V}, I_C = 3\text{ A}$		2.5	V
C_{ob}	Output Capacitance	$V_{CB} = 10\text{ V}, I_E = 0,$ $f = 0.1\text{ MHz}$		200	pF

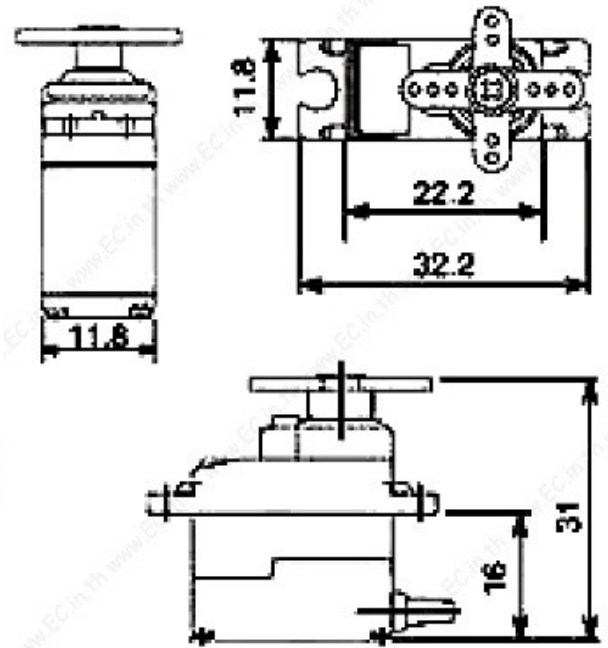
Note:

1. Pulse test: $p_w \leq 300\text{ }\mu\text{s}$, duty cycle $\leq 2\%$.

TIP120 / TIP121 / TIP122 — NPN Epitaxial Darlington Transistor

TIP120 / TIP121 / TIP122 — NPN Epitaxial Darlington Transistor

SG90 9 g Micro Servo

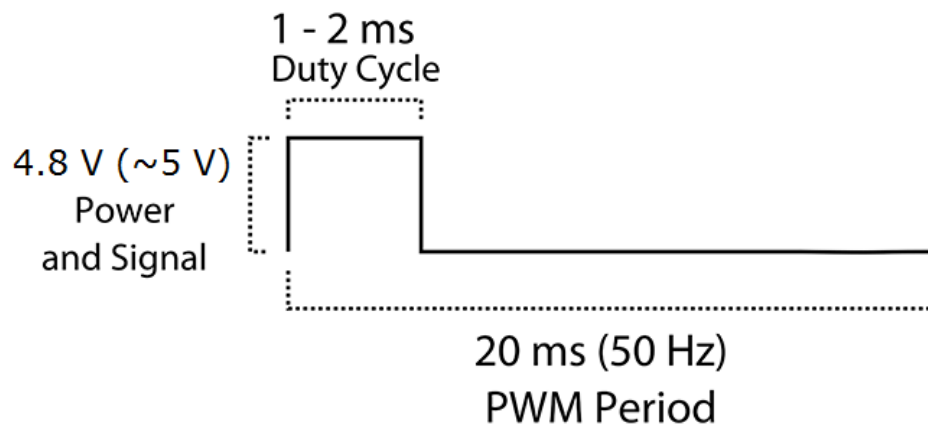
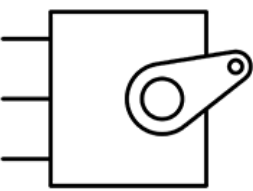


Tiny and lightweight with high output power. Servo can rotate approximately 180 degrees (90 in each direction), and works just like the standard kinds but *smaller*. You can use any servo code, hardware or library to control these servos. Good for beginners who want to make stuff move without building a motor controller with feedback & gear box, especially since it will fit in small places. It comes with a 3 horns (arms) and hardware.

Specifications

- Weight: 9 g
- Dimension: 22.2 x 11.8 x 31 mm approx.
- Stall torque: 1.8 kgf·cm
- Operating speed: 0.1 s/60 degree
- Operating voltage: 4.8 V (~5V)
- Dead band width: 10 μ s
- Temperature range: 0 °C – 55 °C

PWM=Orange ($\square$)
Vcc = Red (+)
Ground=Brown (-)



Position "0" (1.5 ms pulse) is middle, "90" (~2 ms pulse) is all the way to the right, "-90" (~1 ms pulse) is all the way to the left.

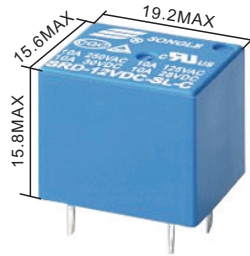
SRD Series SUBMINATURE HIGH POWER RELAY 7A/10A/15A

CUL US
E167996

R50056114
R50137084



CQC08002027667



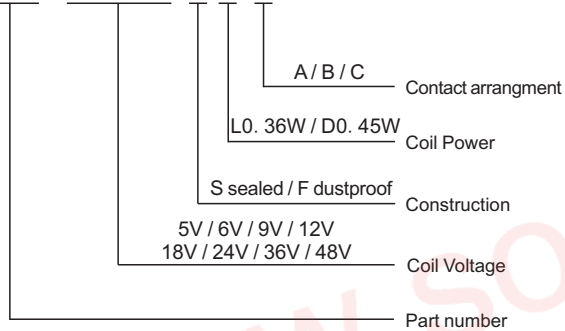
RoHS
Compliant

Features

- Miniature volume, PCB usage
- Contact rating: 7A/10A/15A
- Temperature range: +85°C/+105°C

Ordering Information

SRD-12VDC-S L-C



Contact Rating

Contact arrangement	1A(spsno)/1B(Spsnc)/1C(Spdt)
Contact resistance	100mΩ (1A 6VDC)
Contact material	Silver alloy: AgCdO, AgSnO ₂ , AgNi
Contact rating	7A/250VAC 10A/250VAC 15A/250VAC
Max switching voltage	250VAC
Max switching current	15A
Max switching power	3750VA
Electrical endurance (frequency: 1800 ops/h)	1x10 ⁵ ops / 1x10 ⁴ ops
Mechanical endurance (frequency: 18000 ops/h)	1x10 ⁷ ops

Characteristics

Insulation system		Class B / Class F
Insulation resistance		100MΩ (500VDC)
Dielectric Strength Leakage current 1mA	Between contacts and coil	1500VAC 1 minute
	Between open contacts	1000VAC 1 minute
Operate time(Under nominal voltage)		≤10ms
Release time(Under nominal voltage)		≤10ms
Humidity		85% RH (20℃)
Ambient Temperature		-40℃~+85℃/-40℃~+105℃
Shock resistance	Functional	98m/s ²
	Destructive	980m/s ²
Vibration resistance (Double amplitude)		10Hz~55Hz 1.5mm
Weight		Approx 9g
Construction		Sealed

Remark: 1) All above data is initial value

Coil Data

(at 20°C)

Power dissipation W	Voltage VDC	Current mA	Resistance Ω ± 10%	Pick-up voltage	Drop-out voltage	Max allowable
0.36W (L)	05	71.4	70	75%Max	10%Min	130%
	06	60	100			
	09	40	225			
	12	30	400			
	18	20	900			
	24	15	1600			
	36	10	3600			
	48	7.5	6400			
0.45W (D)	05	89.3	55	75%Max	10%Min	130%
	06	75	80			
	09	50	180			
	12	37.5	320			
	18	25	720			
	24	18.7	1280			
	36	12.5	2880			
	48	10	4500			

This product specification only for your reference, we will not notify you if we do any improvement. We reserve all the right for the final explanation



Web: www.songle.com www.songlerelay.com

Add.: CW7, Zone A, Far East Industrial Park, Yuyao City, Zhejiang, China Zip Code: 315400

SRD Series SUBMINATURE HIGH POWER RELAY 7A/10A/15A

Safety Approval Ratings

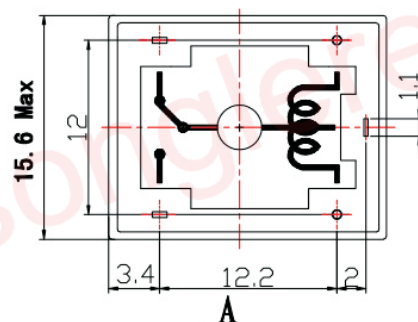
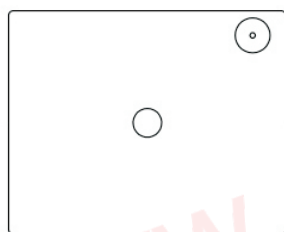
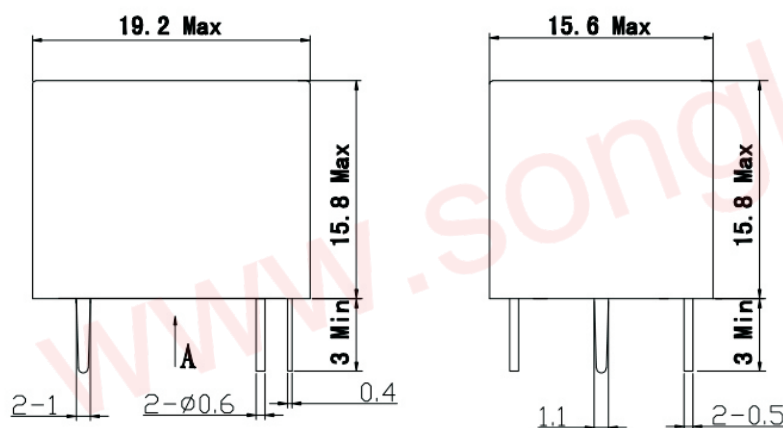
CQC CQC08002027667
7A 250VAC 10A 250VAC

△ R50056114 R50137084
7A 250VAC 10A 250VAC 15A 250VAC NO

UL US E167996
3A 125VAC inductive 10A 125VAC 15A 125VAC NO

OUTLINE DIMENSIONS ,WIRING DIAGRAM AND PC BOARD LAYOUT Unit:mm

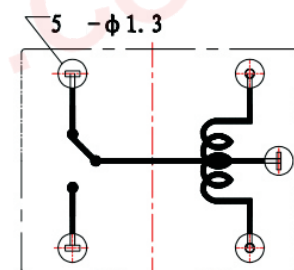
Physical installation diagram



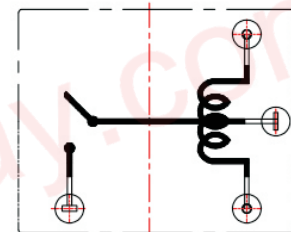
(Bottom view)

Tolerance: $\pm 0.1\text{mm}$

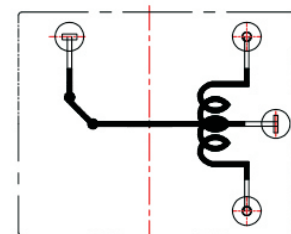
PCB board diagram
(bottom view diagram)



Conversion C type



Normally open A type



Normally closed B type

This product specification only for your reference, we will not notify you if we do any improvement. We reserve all the right for the final explanation